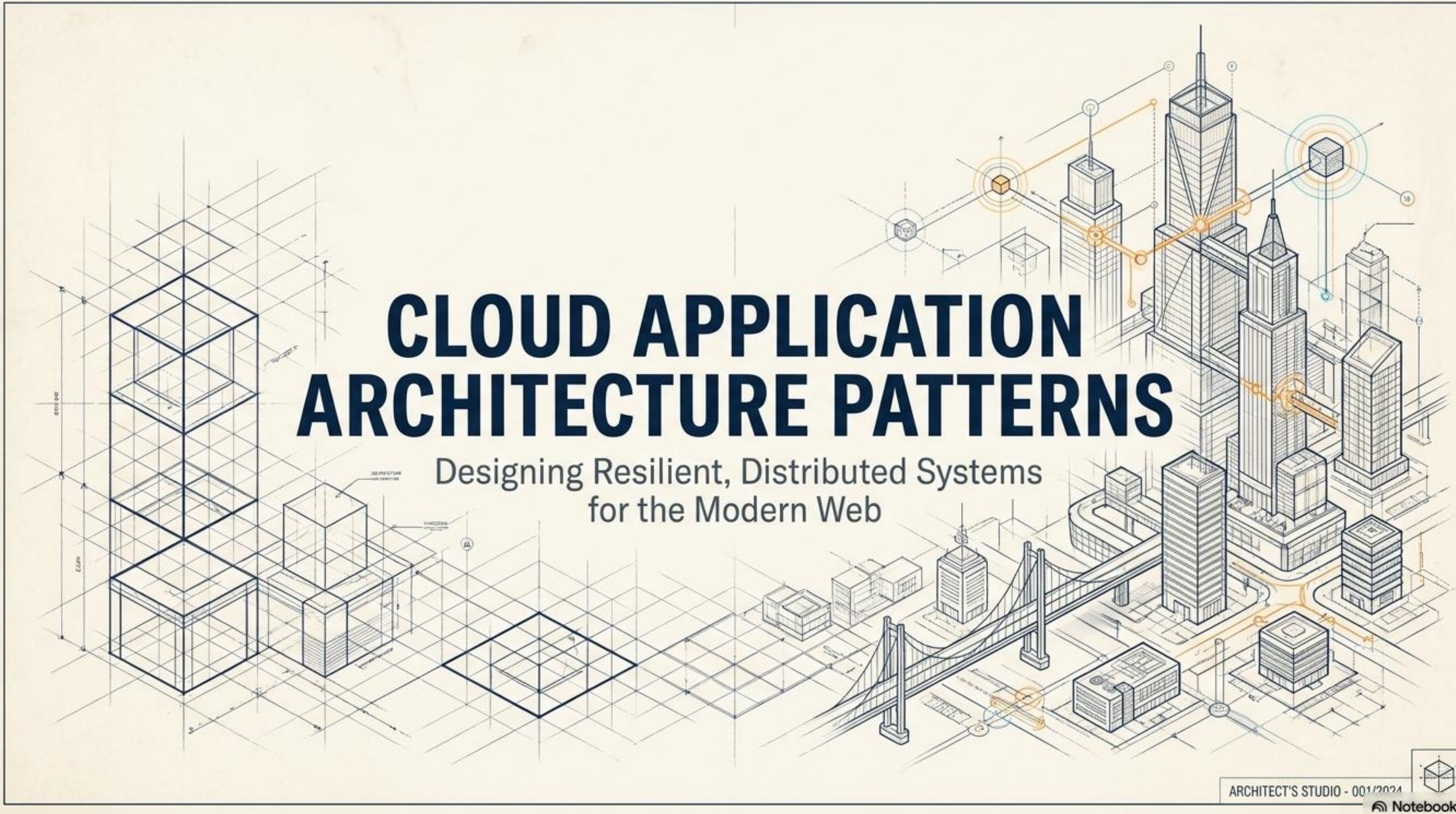

CLOUD APPLICATION ARCHITECTURE PATTERNS
Designing Resilient, Distributed Systems for the Modern Web
ARCHITECT'S STUDIO - 001/2024
NotebookLM

O'REILLY

Cloud Application Architecture Patterns

Designing, Building, and Modernizing for the Cloud

Kyle Brown, Bobby Woolf & Joseph Yoder
Foreword by Sam Newman

THE PENDULUM OF APPLICATION ARCHITECTURE

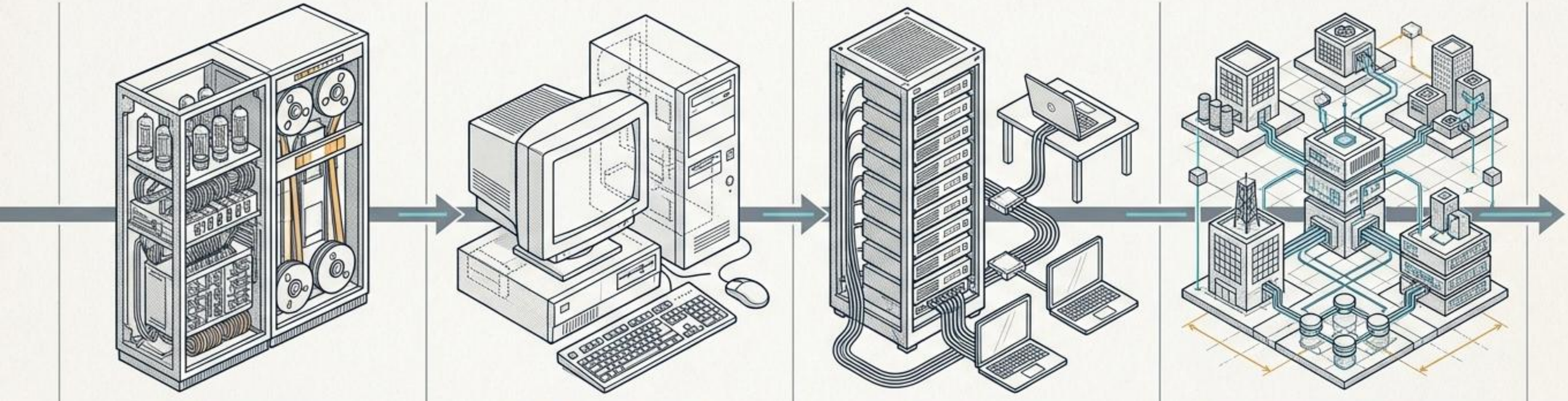


FIG. 1 - MAINFRAME UNIT

FIG. 2 - PERSONAL COMPUTER

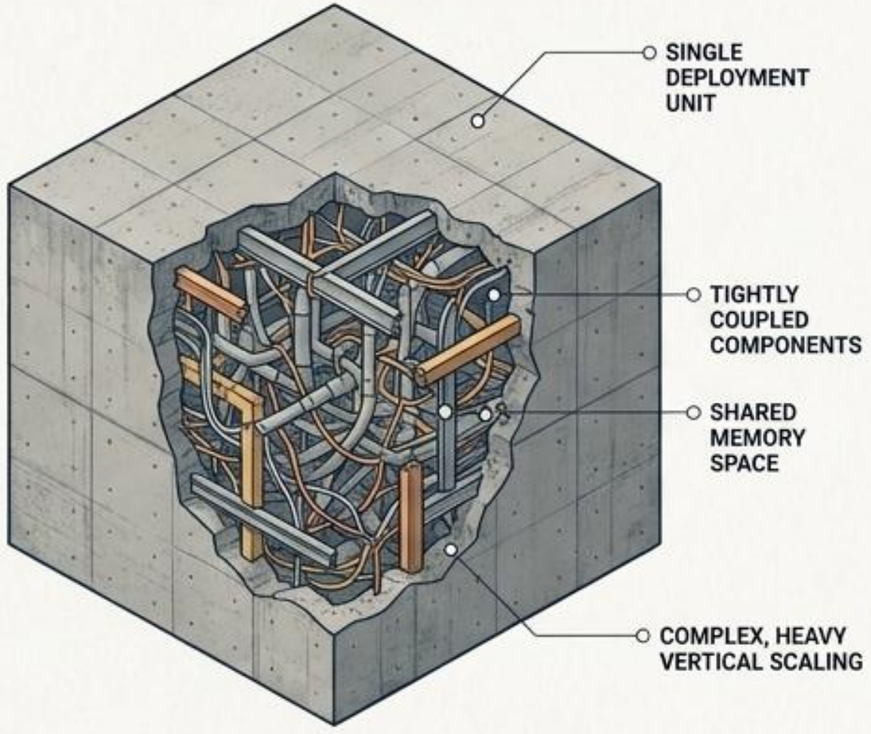
FIG. 3 - CLIENT/SERVER NETWORK

FIG. 4 - CLOUD-NATIVE INFRASTRUCTURE

MAINFRAME	DESKTOP	CLIENT/SERVER	CLOUD-NATIVE
Centralized compute, terminal access.	Local execution, siloed data.	Shared database, heavy client logic.	Distributed, API-driven, elastic scaling.

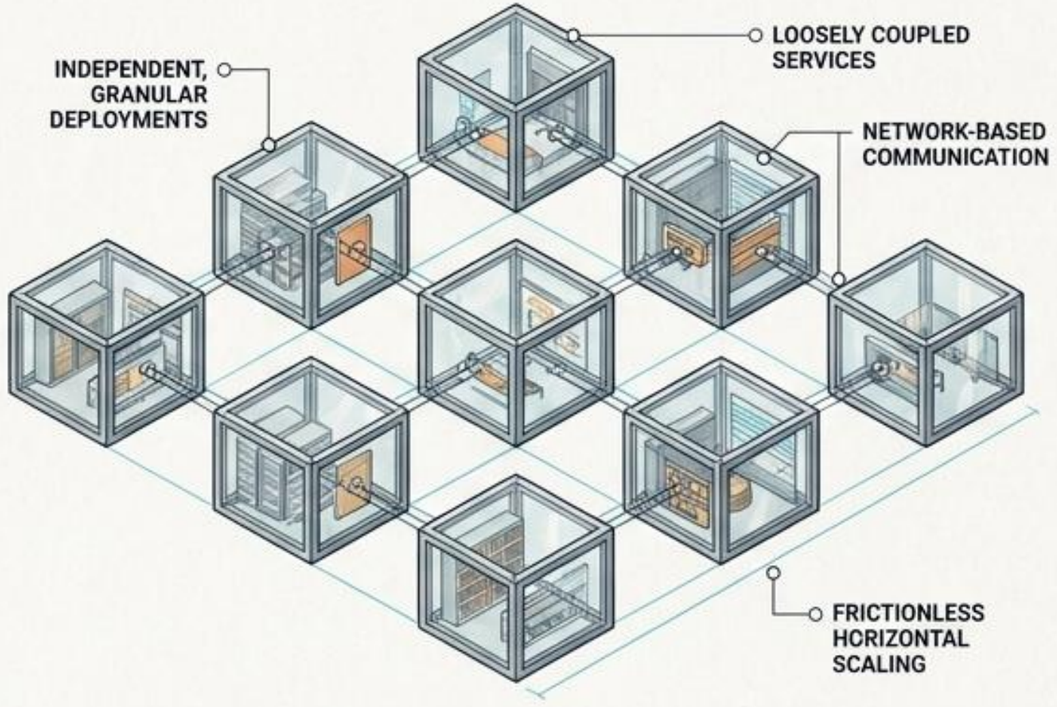
THE ARCHITECTURAL SHIFT: MONOLITHS VS. DISTRIBUTED SYSTEMS

MONOLITHIC ARCHITECTURE



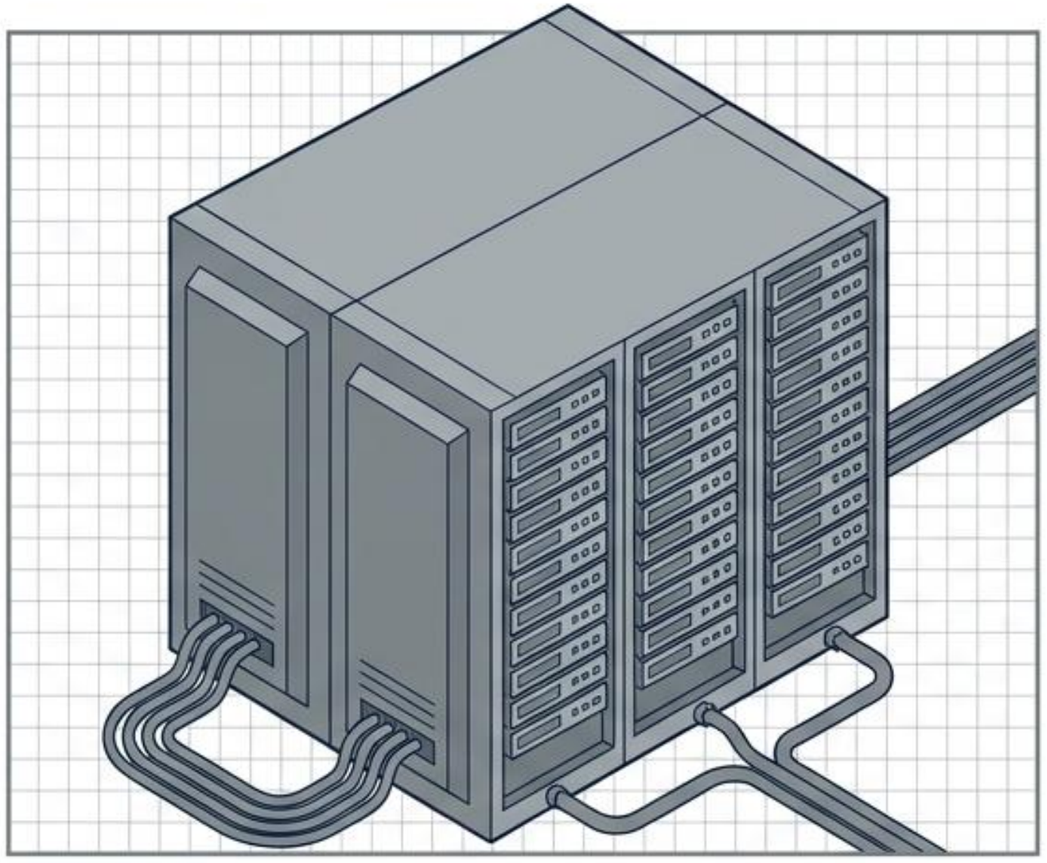
Single deployment unit
Tightly coupled components
Shared memory space
Complex, heavy vertical scaling

DISTRIBUTED ARCHITECTURE



Independent, granular deployments
Loosely coupled services
Network-based communication
Frictionless horizontal scaling

THE PHYSICAL LAWS OF INFRASTRUCTURE HAVE FUNDAMENTALLY CHANGED



ON-PREMISES

Rigid. Capital Intensive. Location Dependent.

CLOUD NATIVE

Elastic. Operational Expense. Location Agnostic.

EPHEMERAL INFRASTRUCTURE

Servers are now cattle, not pets. They vanish without warning.

NETWORK UNRELIABILITY

Latency spikes and dropped packets are guaranteed constants.

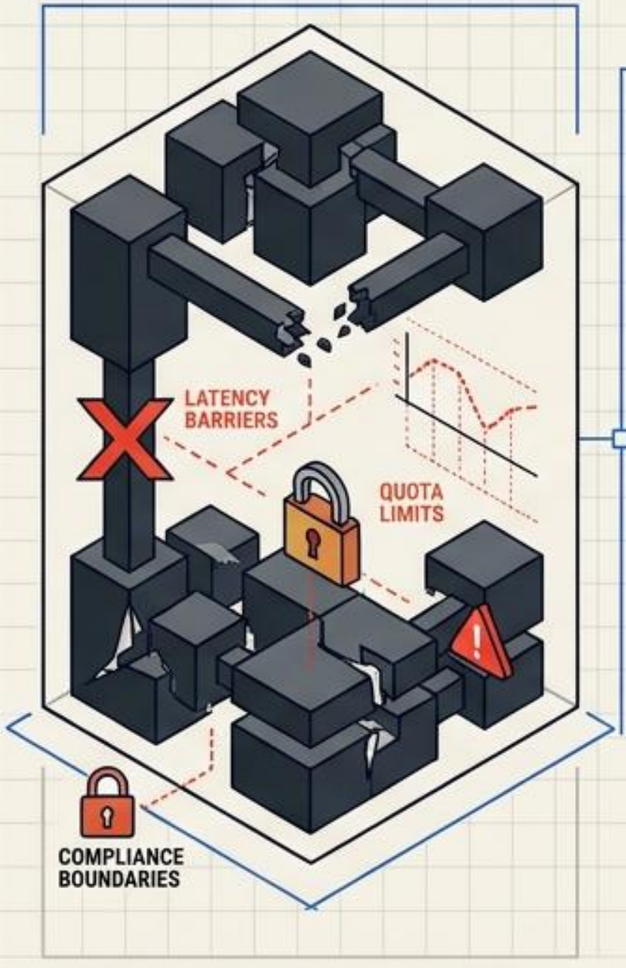
HORIZONTAL SCALING

Growth requires adding more nodes, not buying bigger machines.

Cloud Architecture Patterns

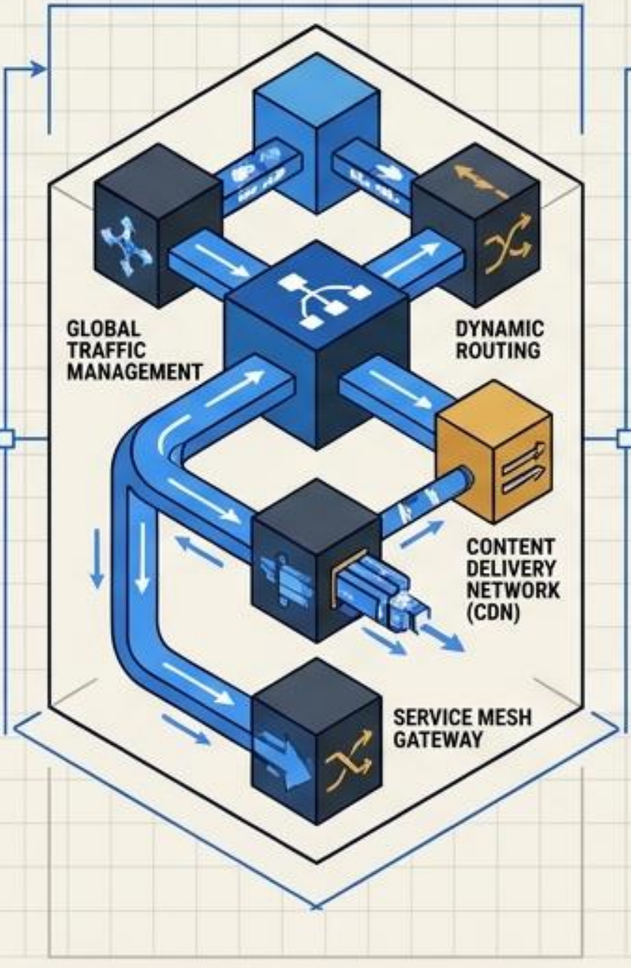
Engineering Resilience and Scale in Distributed Systems

1. Cloud Constraints



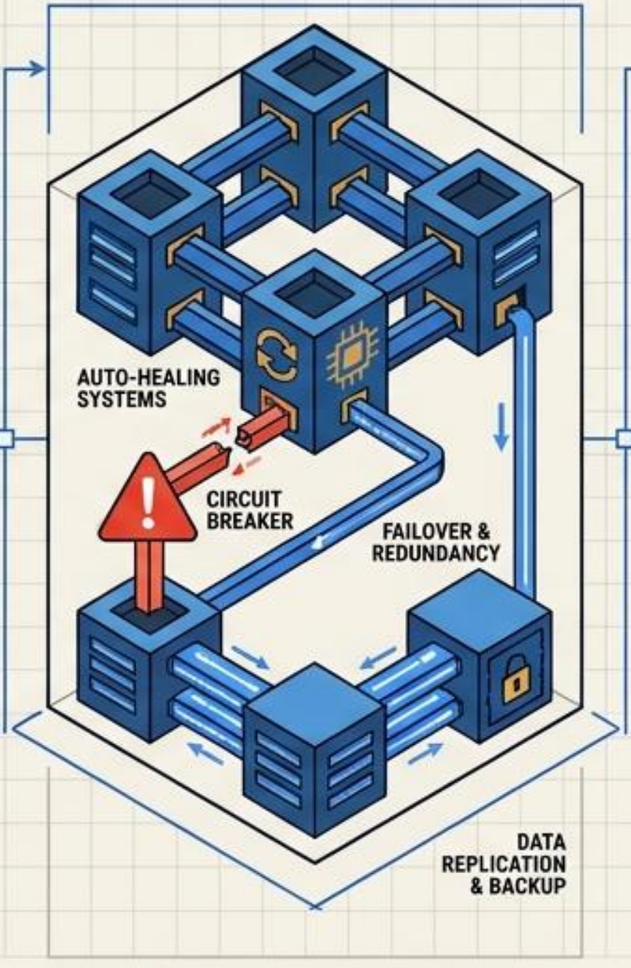
Limitations impacting system design; requires careful consideration of resource caps and geographical restrictions.

2. Access Routing



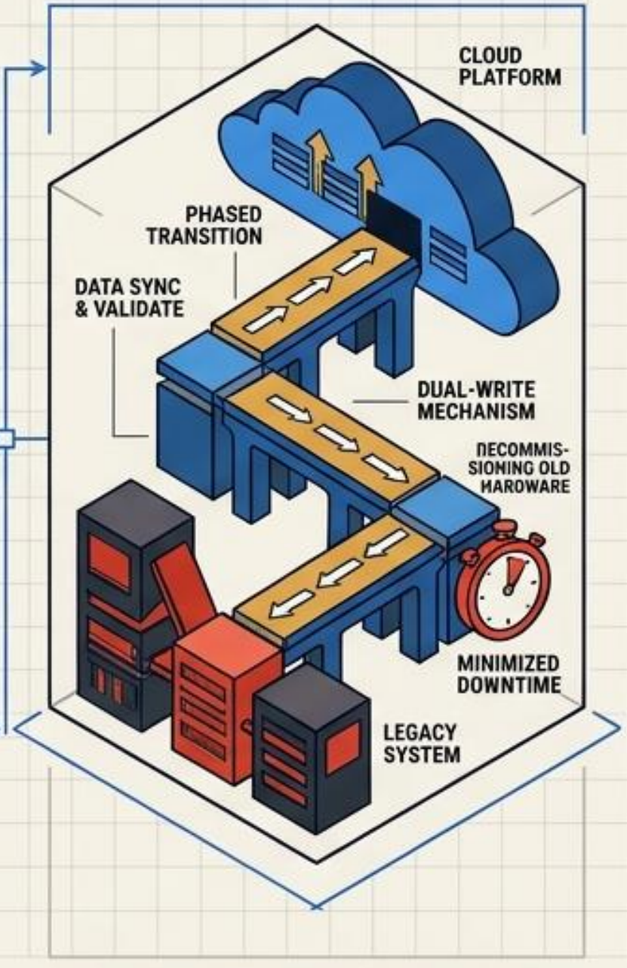
Directing requests efficiently to appropriate resources; optimizes performance and ensures secure entry points.

3. Resiliency Engineering



Building fault-tolerant systems that recover from failures; focuses on redundancy and automated recovery strategies.

4. Data & Incremental Migration



Moving data and workloads gradually to the cloud; reduces risk and ensures business continuity during the shift.

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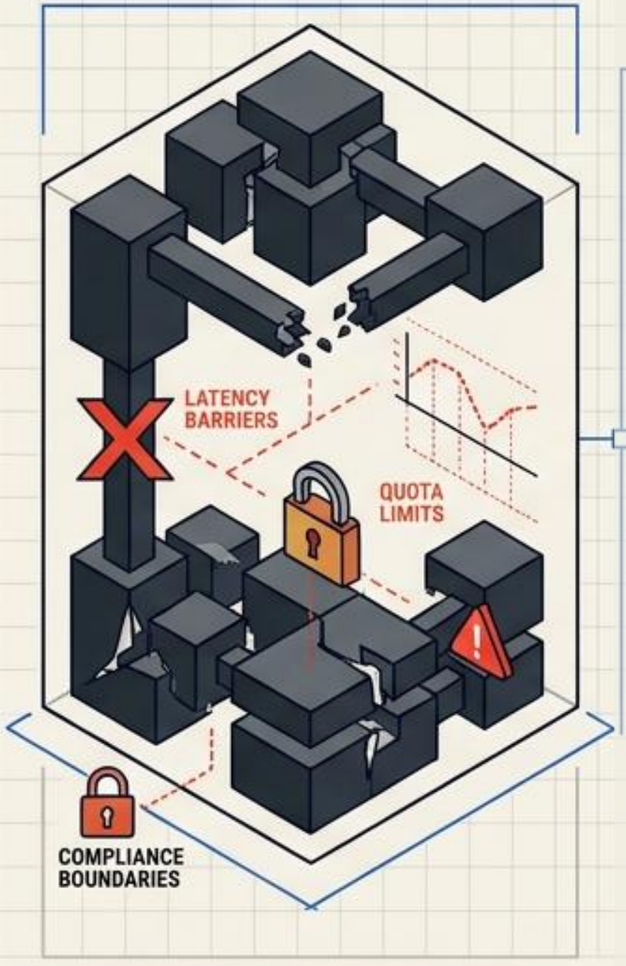

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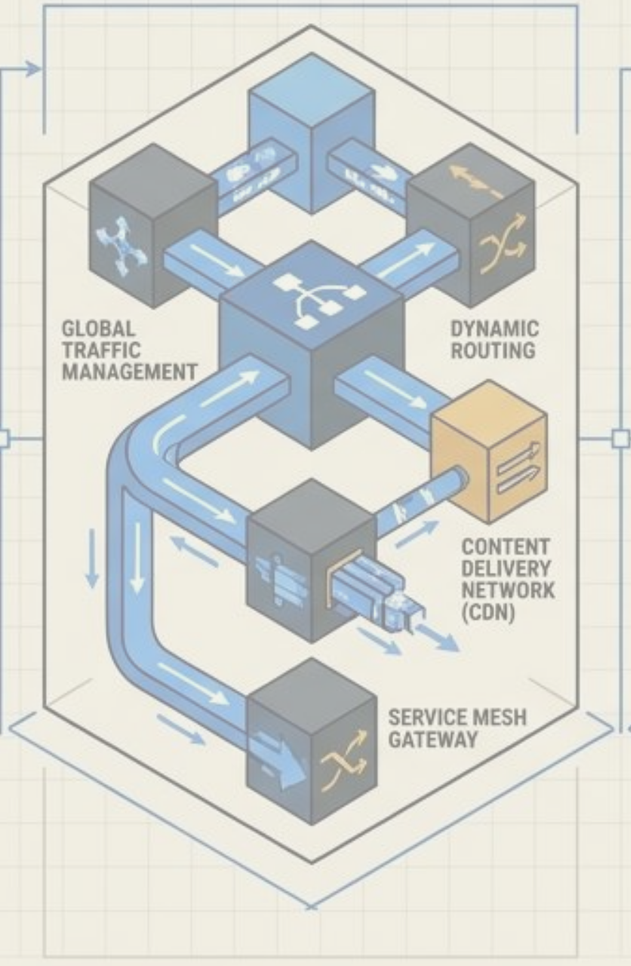

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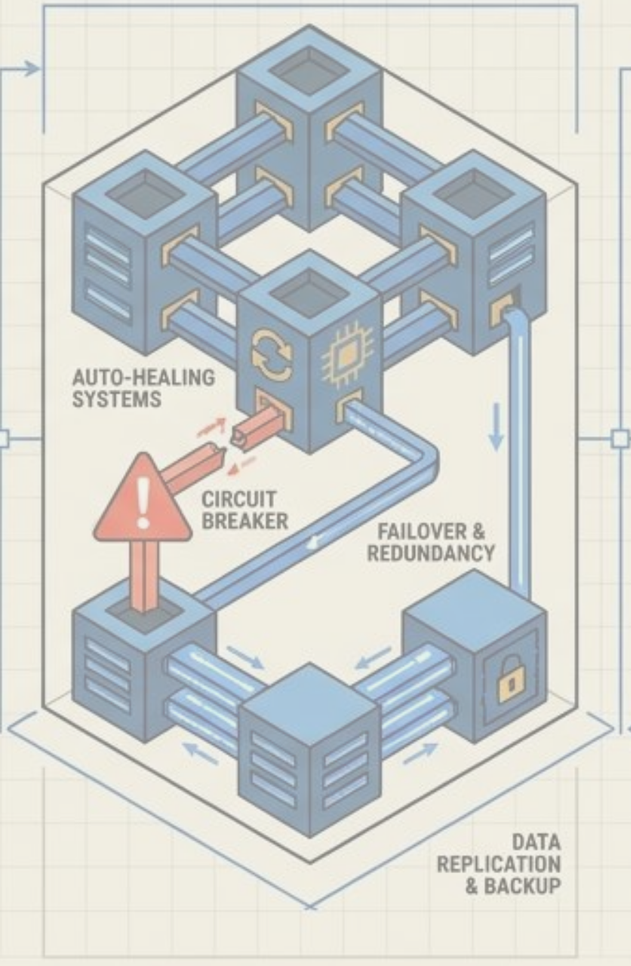

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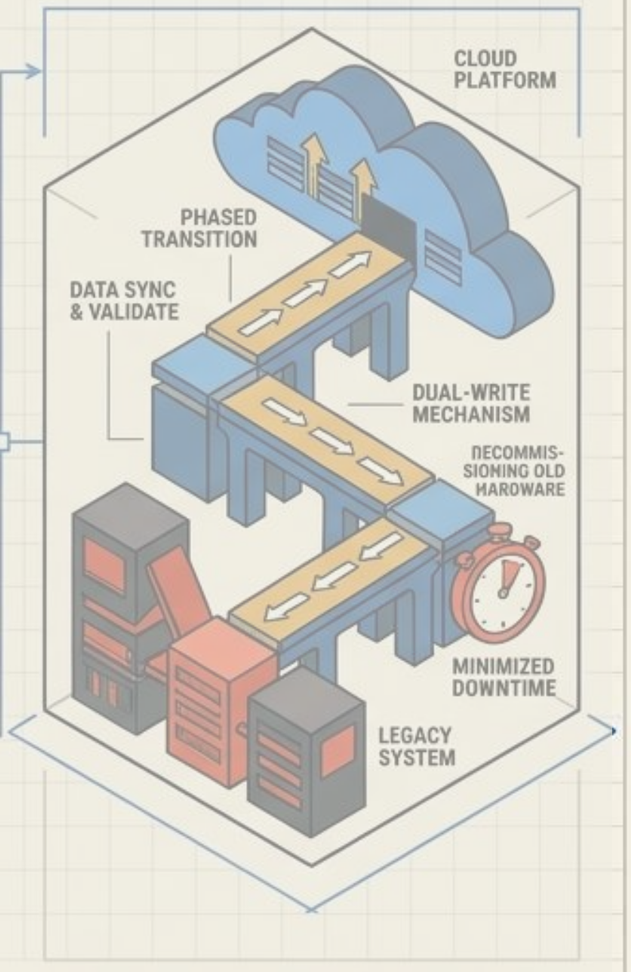

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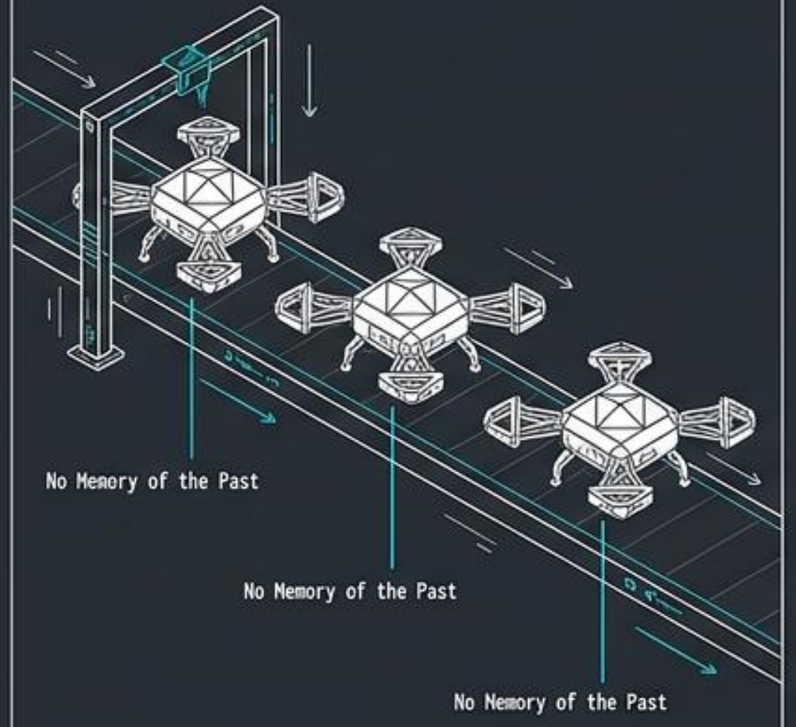
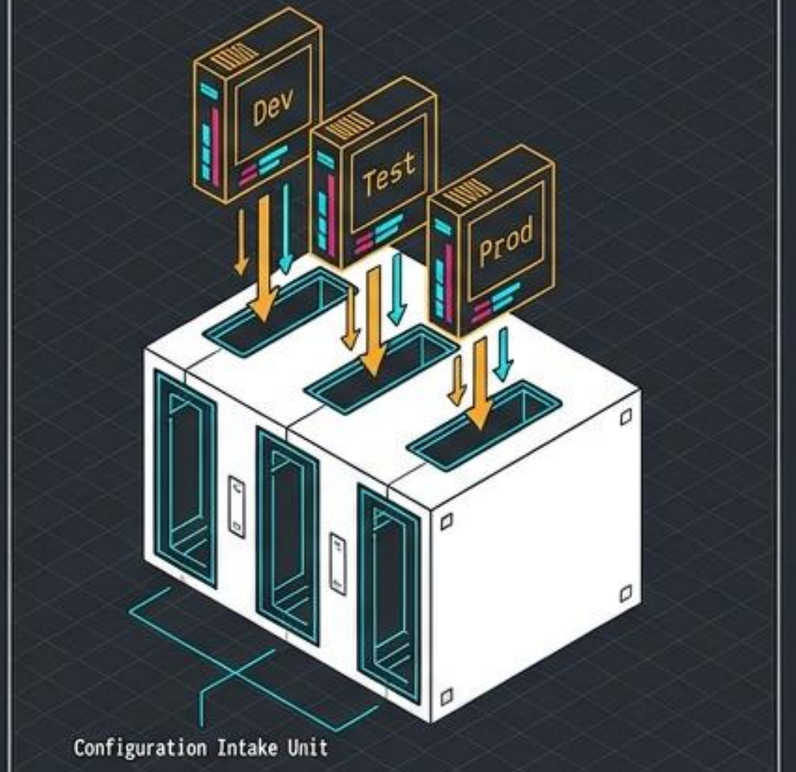
Cloud-Native Foundations: The Zoning Laws

Cloud environments demand ephemeral infrastructure governed by strict zoning laws.

Stateless & Replicable



External Configuration



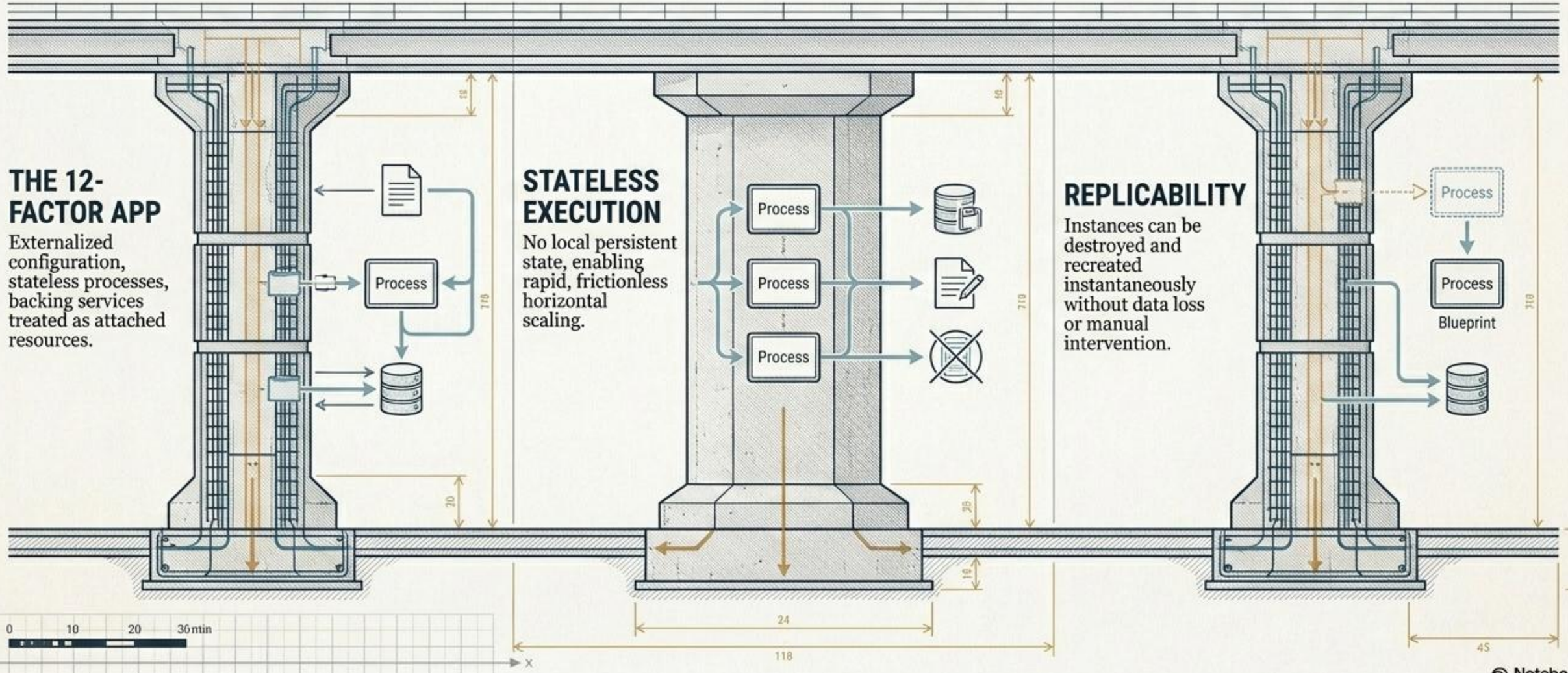
Ephemeral Infrastructure

Dismantled "Pet" Server: Complex, Singte Peint of Failure

Replicable "Cattle" Grid: Scalable, Disposabled, Standardized

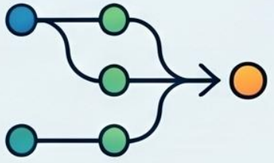
Architectural Trade-Off Scale

THE STRUCTURAL PILLARS OF CLOUD-NATIVE APPLICATIONS



THE TWELVE-FACTOR APP: PRINCIPLES FOR CLOUD-NATIVE SUCCESS

A high-level reference for building scalable, resilient, and portable cloud applications through automation and environmental parity.



1. Codebase

One tracked codebase, multiple deployments.

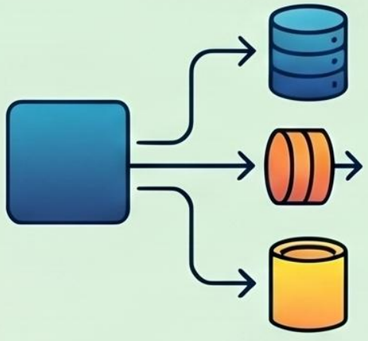
2. Dependencies

Explicitly declare and isolate all dependencies.

3. Config

Store configurations in the environment, not code.

FOUNDATIONS & ENVIRONMENT



4. Backing Services

Treat databases/queues as interchangeable attached resources.

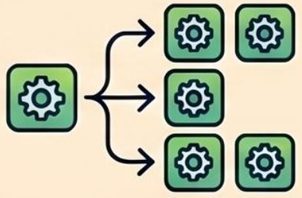
EXECUTION & SCALABILITY

5. Build, Release, Run

Strictly separate build, release, and run stages.

6. Processes

Execute apps as stateless, share-nothing processes.

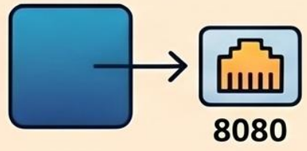


8. Concurrency

Scale out by adding more independent processes.

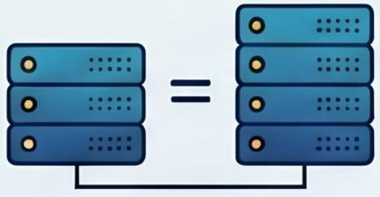
9. Disposability

Maximize robustness with fast startup and graceful shutdown.



7. Port Binding

Export services directly via port binding.



10. Dev/Prod Parity

Keep development and production environments as similar as possible.

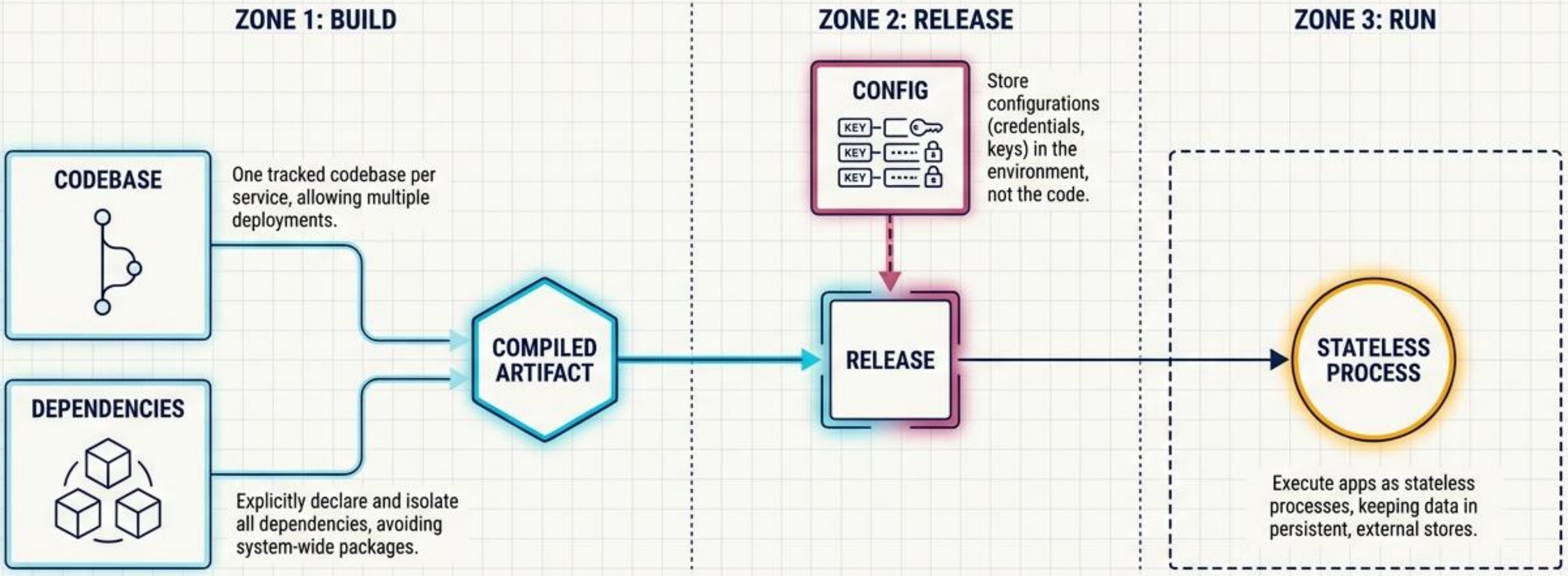
11. Logs

Treat logs as continuous, unbuffered event streams.

12. Admin Processes

Run management tasks (e.g., migrations) as one-off processes.

DELIVERY: THE CLOUD-NATIVE PIPELINE



TECHNICAL RATIONALE

According to the Twelve-Factor methodology, a cloud-native service must start with a single, tracked codebase to allow multiple, reliable deployments. Furthermore, we must explicitly declare and isolate dependencies to avoid system-wide package conflicts.

As we move to the Release phase, we inject Configuration. The technical rationale here is strict separation: credentials and environment-specific keys must live in the environment, not the code. This ensures the exact same build artifact can be deployed anywhere safely.

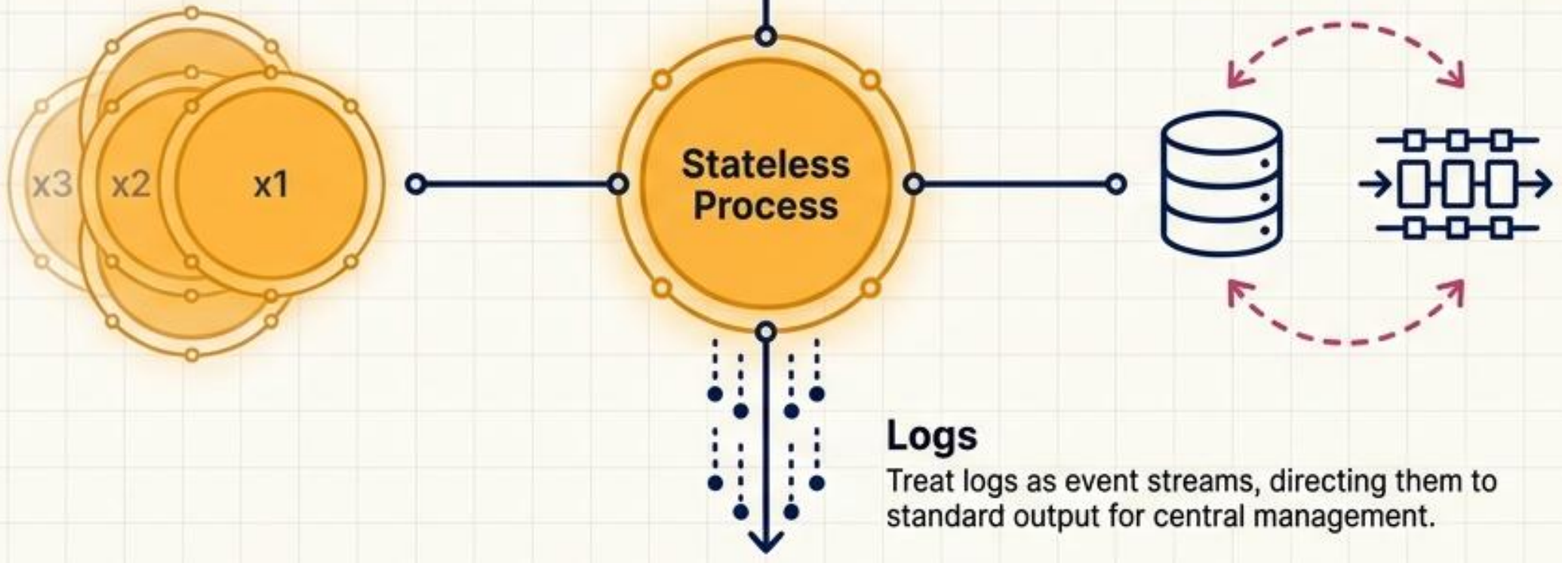
Notice the strict separation of stages. This prevents runtime mutations and ensures predictability. Finally, the app boots into the Run phase as a completely stateless process. By design, any persistent data must be pushed to external stores, which guarantees our app can be killed or replicated at a moment's notice.

Operations: The Service Anatomy

Port Binding
Self-contain services by exporting them via port binding.

Concurrency
Scale out by adding more processes, rather than increasing capacity per process.

Backing Services
Treat databases, queues, and caches as attached resources, interchangeable via config.



Logs
Treat logs as event streams, directing them to standard output for central management.

Technical Rationale

Building on our pipeline, the Stateless Process sits at the center of our service anatomy. Because it holds no local state, it acts as a lightweight, interchangeable engine.

To operate in the cloud, it must be completely self-contained, which is achieved by exporting itself itself via Port Binding. It doesn't rely on a host web server.

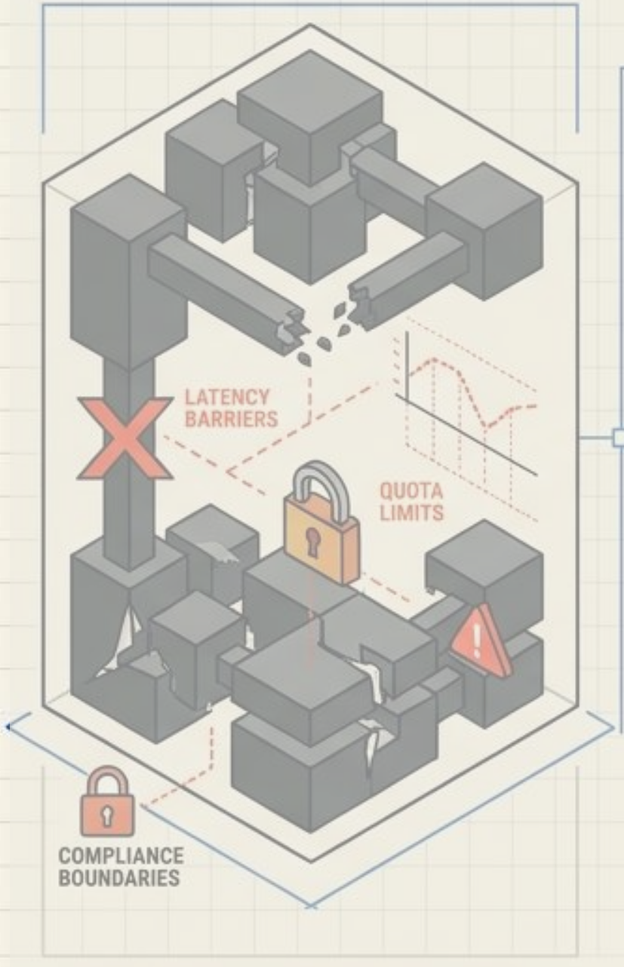
Databases or message queues are treated merely as "Backing Services"—attached resources that can be swapped out via config without changing the app's code. Additionally, because our hub is stateless, we achieve true Concurrency. We scale horizontally by adding more identical processes to handle the load.

Notice the outward flow of Logs. In a distributed cloud environment, local log files are an anti-pattern. We treat logs strictly as event streams pushed to standard output, allowing centralized systems to aggregate and manage them.

Cloud Architecture Patterns

Engineering Resilience and Scale in Distributed Systems

1. Cloud Constraints



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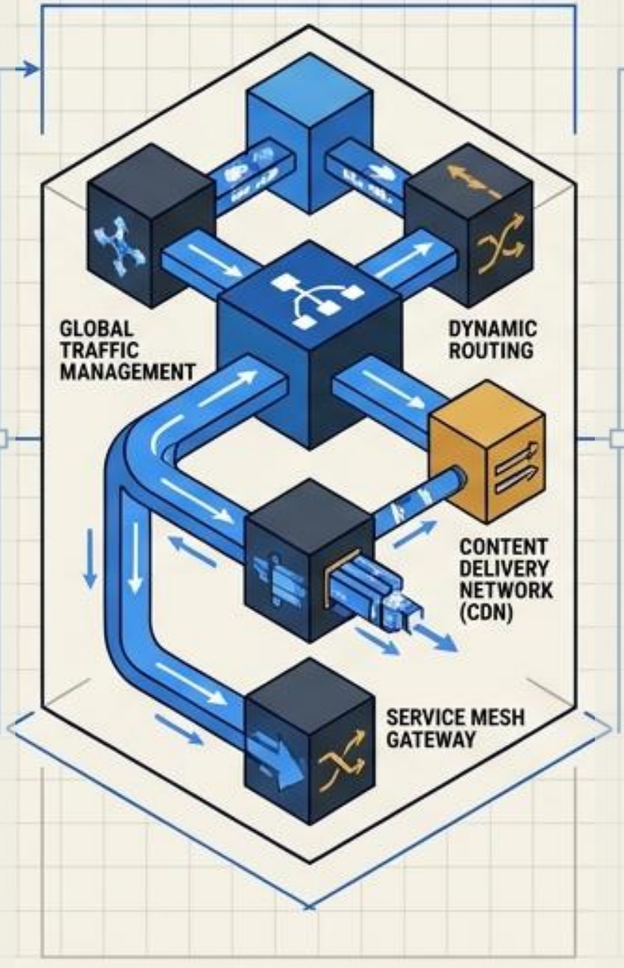
2. Access Routing



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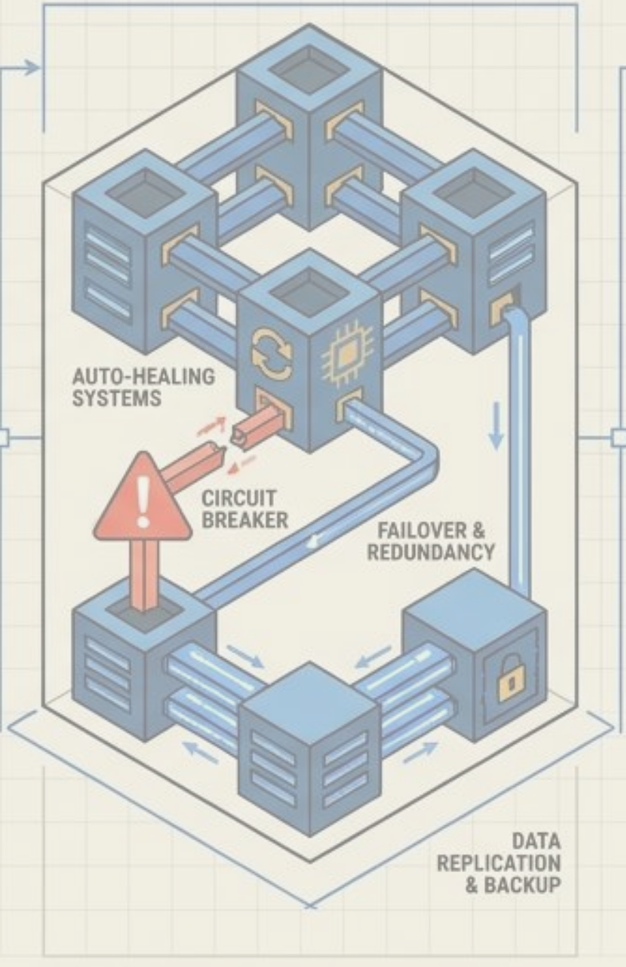

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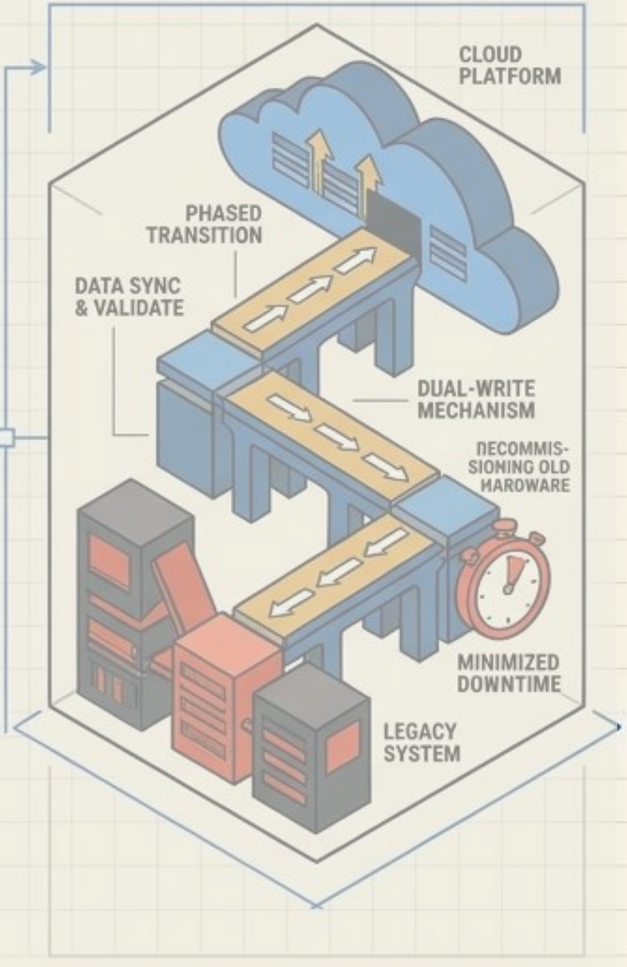
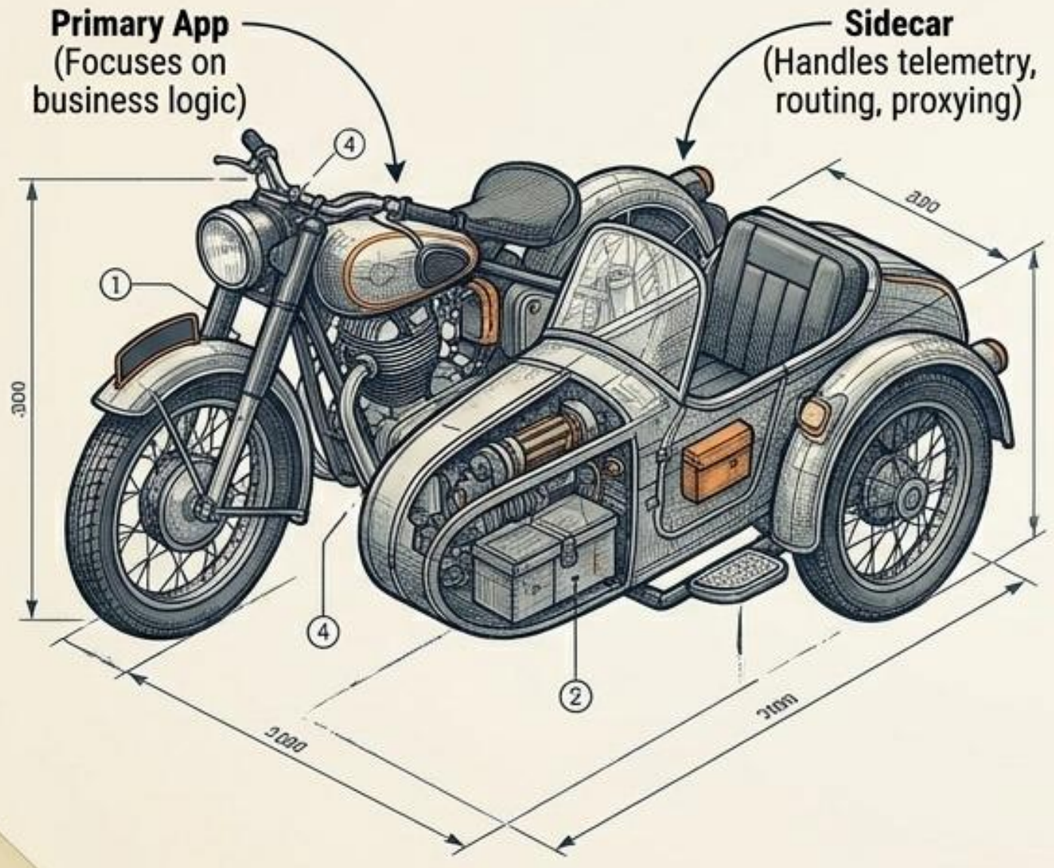


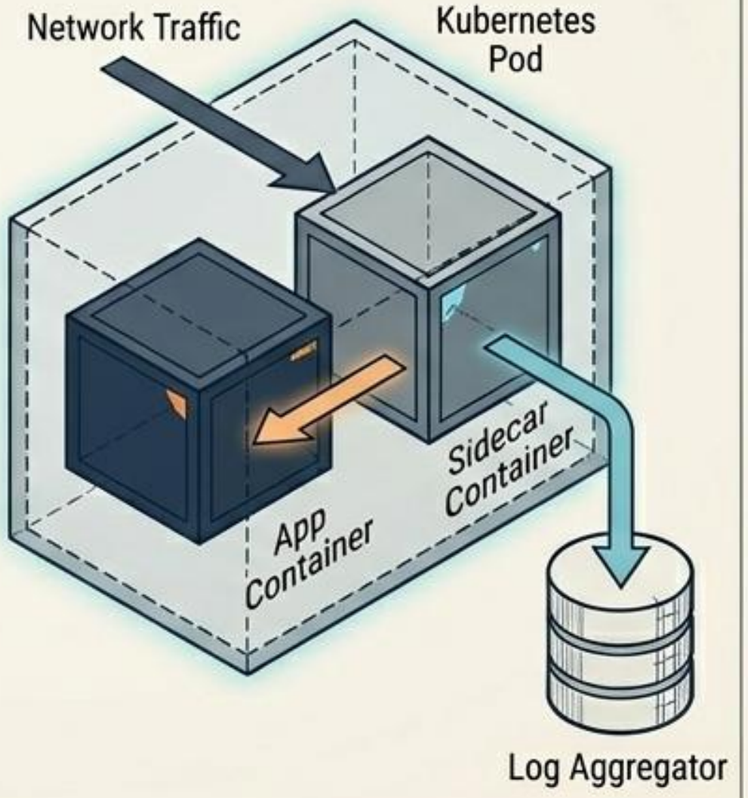
Moving data and workloads gradually to the cloud; reduces risk and ensures business continuity during the shift.

PATTERN 1: SIDECAR (MICROSERVICES)

THE ANALOGY



TECHNICAL FLOW



ARCHITECTURAL TRADE-OFFS

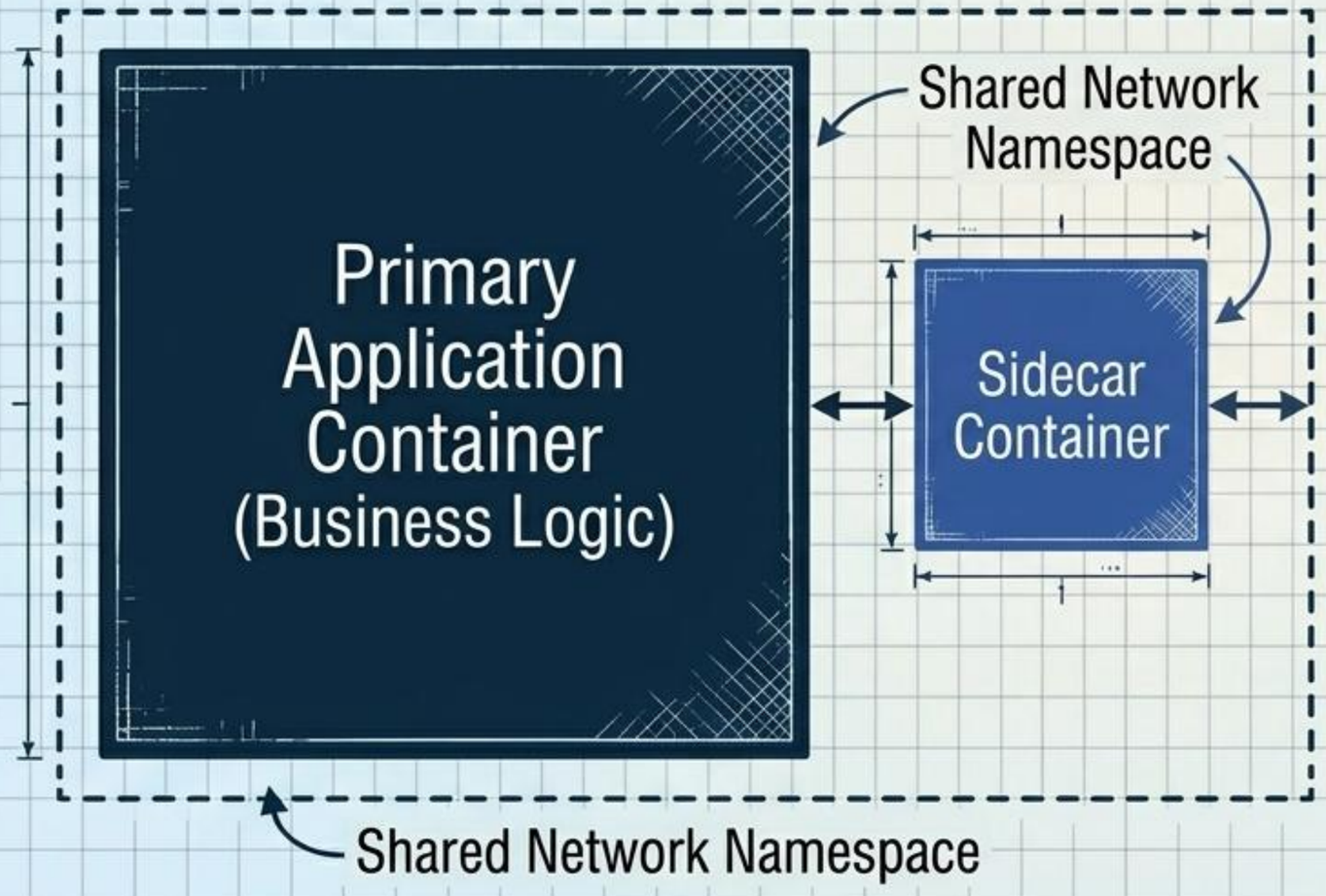
⊕	Abstracts infrastructure complexity
⊕	Language agnostic
⊕	Independent updates
⊖	(-) Increases deployment complexity
⊖	(-) Slight network latency
⊖	(-) Resource overhead

SCALE 1:100

PROJ. REF: ARCH-2024 | DATE: OCT 26

OFFLOADING OPERATIONAL WEIGHT WITH SIDECARS

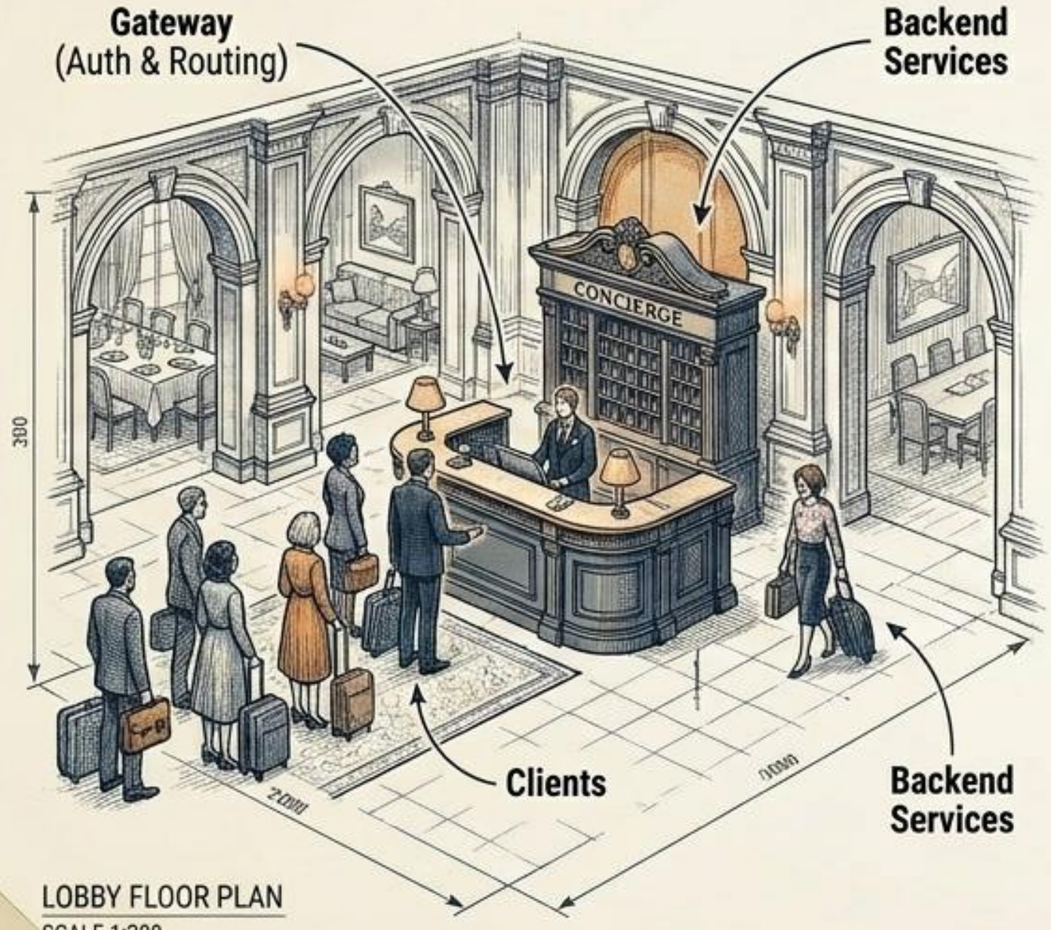
A structural container diagram that clearly illustrates physical coupling and logical boundaries. The slide uses geometric hierarchy to show main components versus helper components.



- Deploys helper components alongside the primary application container.
- Handles cross-cutting concerns: Logging, Monitoring, Security Proxying.
- Shares the exact same lifecycle and network namespace.
- Keeps primary application code focused purely on business logic.

PATTERN 2: API GATEWAY (CLOUD CLIENTS)

THE ANALOGY



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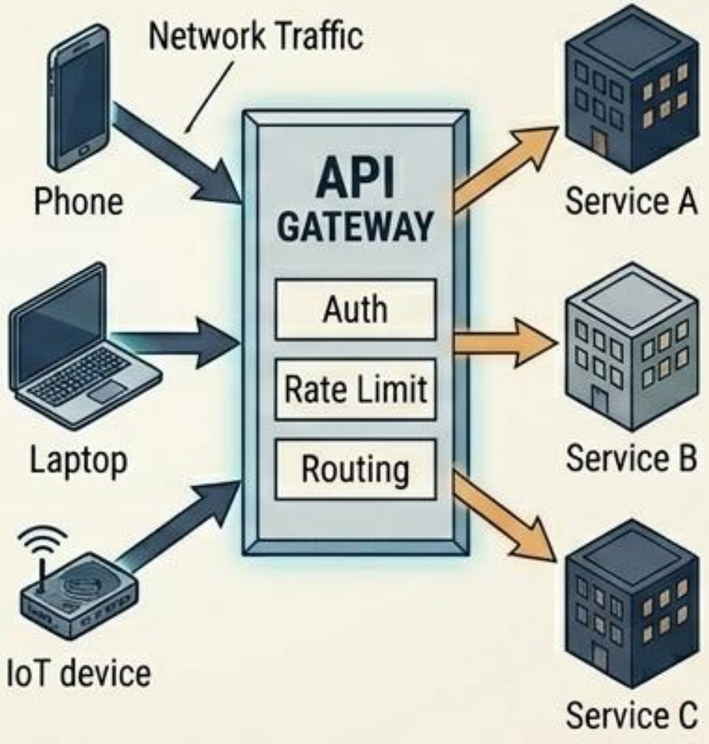


ARCHITECTURAL TRADE-OFFS

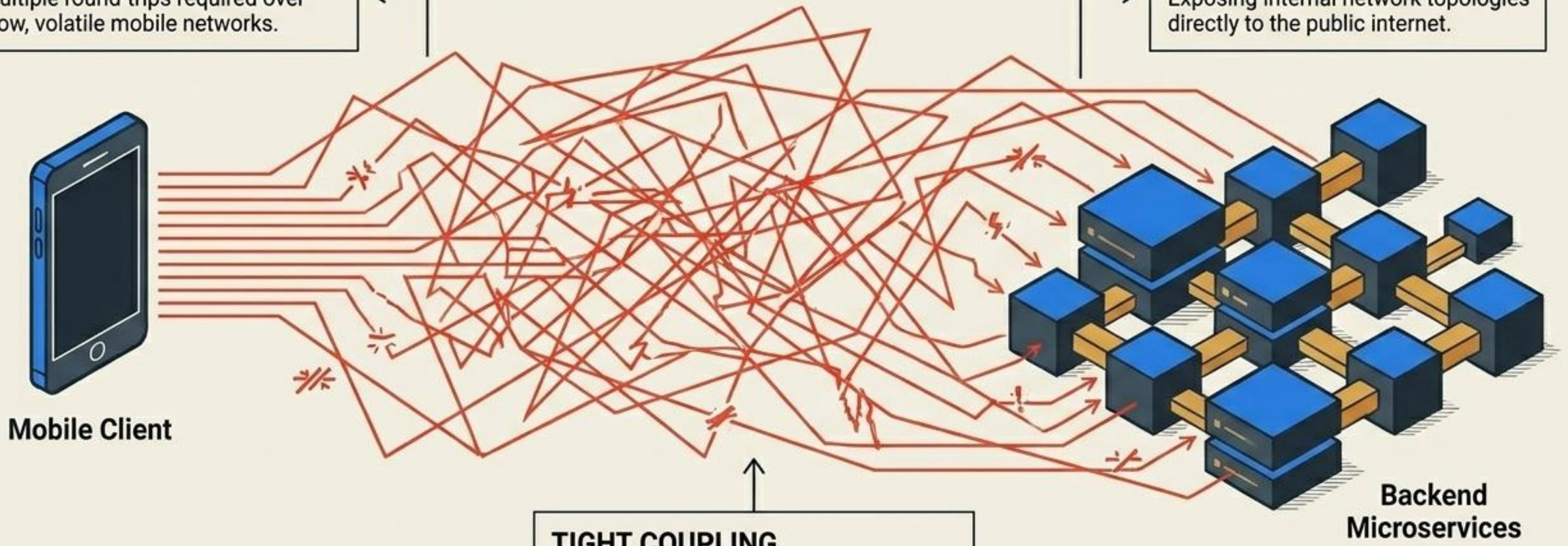
⊕	(+) Single client entry point
⊕	(+) Centralized security policies
⊕	(+) Internal protocol translation
⊖	(-) Single point of failure
⊖	(-) Potential network chokepoint
⊖	(-) Risk of business logic bloating

DIRECT CLIENT CONNECTIONS CREATE CASCADING CHAOS

The "Chatty Client" Anti-Pattern

CRIPPLING LATENCY
Multiple round-trips required over slow, volatile mobile networks.

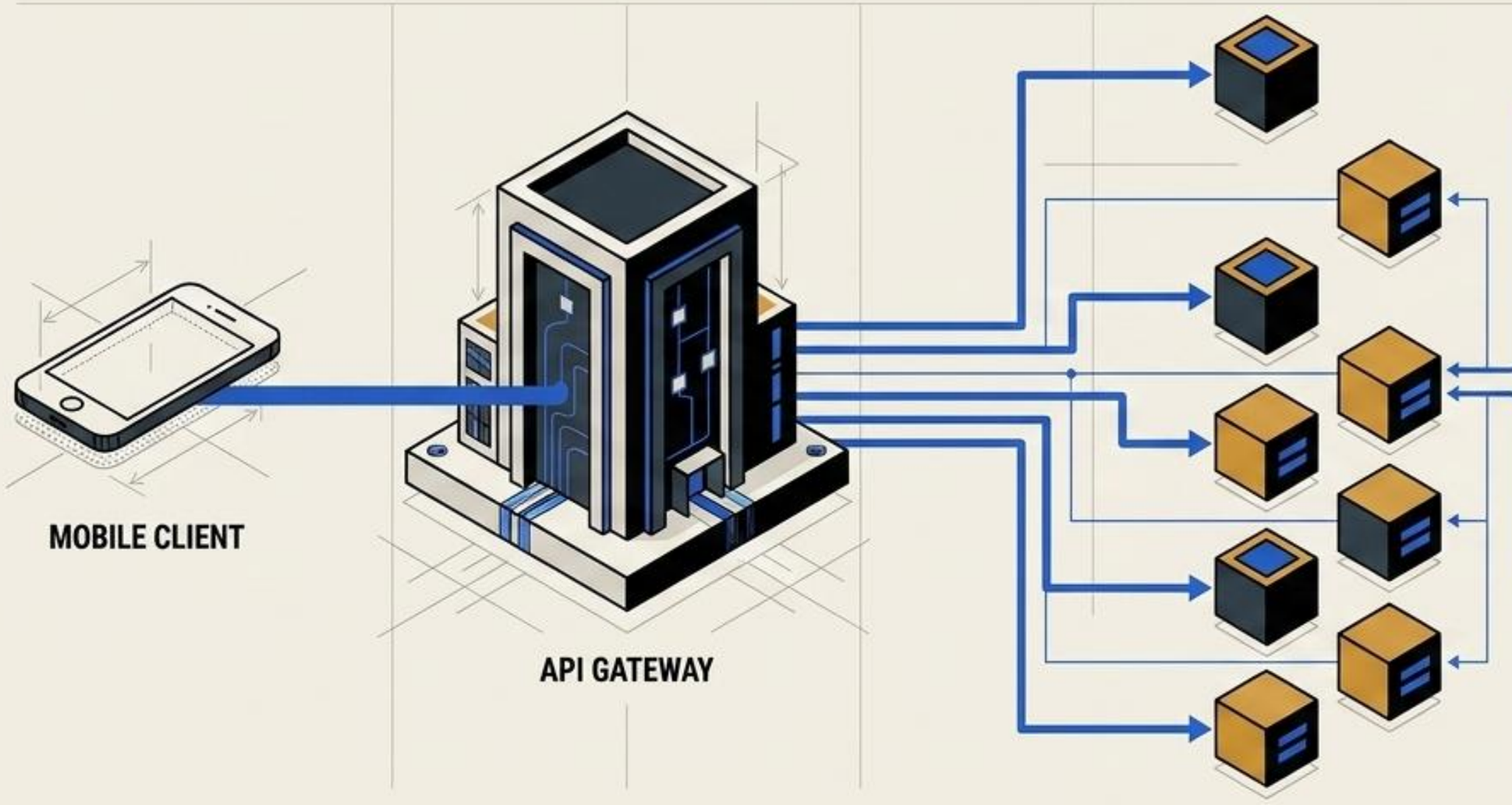
SECURITY NIGHTMARES
Exposing internal network topologies directly to the public internet.



TIGHT COUPLING
Frontend code breaks instantly if backend routing or service boundaries change.

THE API GATEWAY BUFFERS COMPLEXITY FROM THE CLIENT

Definition: A single point of entry acting as a reverse proxy to route client requests to backend services.



INTELLIGENT ROUTING & AGGREGATION

Intelligent routing & aisquation and aggregation acs to route client requests to backend services.

RATE LIMITING & THROTTLING

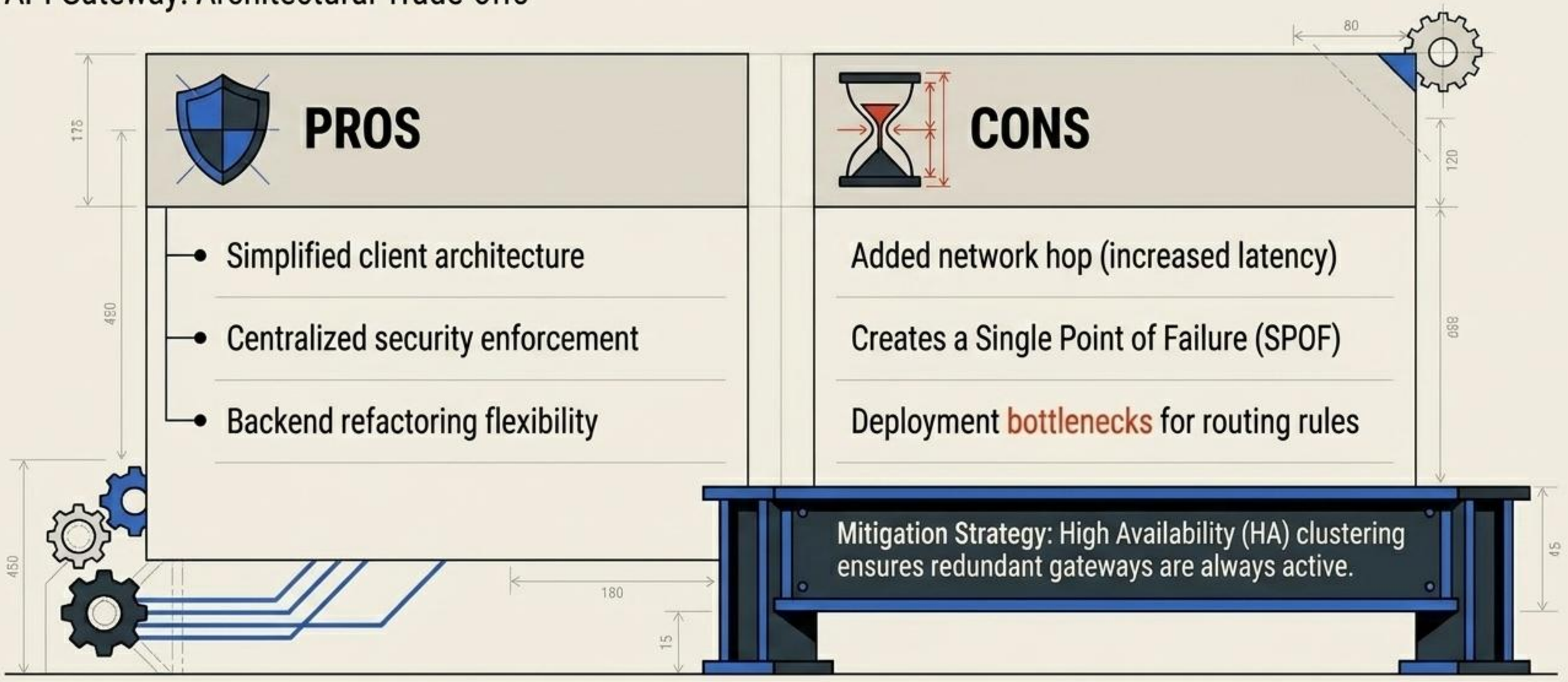
Rate limiting & throttling acting as a reverse proxy to route client requests to/a backend services.

AUTHENTICATION OFFLOADING

Authentication offloading to rams and connectis to route client requests to backend services.

CENTRALIZED ACCESS INTRODUCES NEW DEPLOYMENT BOTTLENECKS

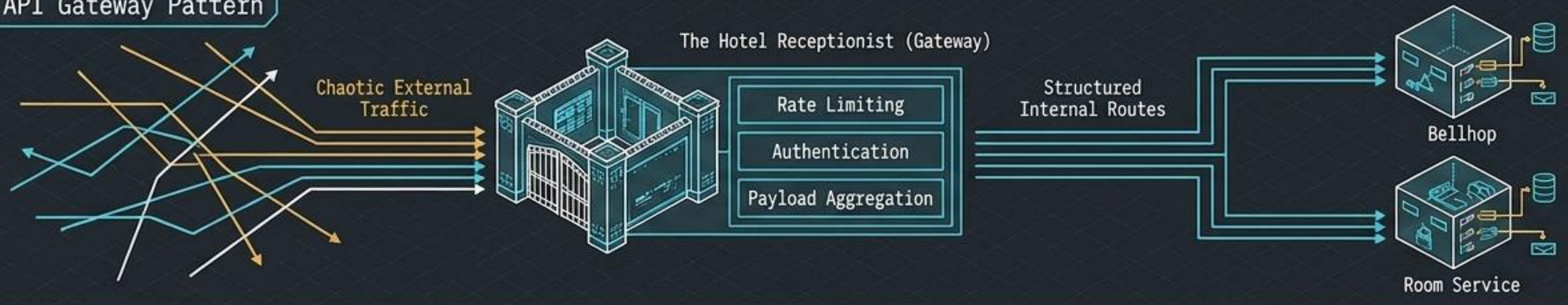
API Gateway: Architectural Trade-offs



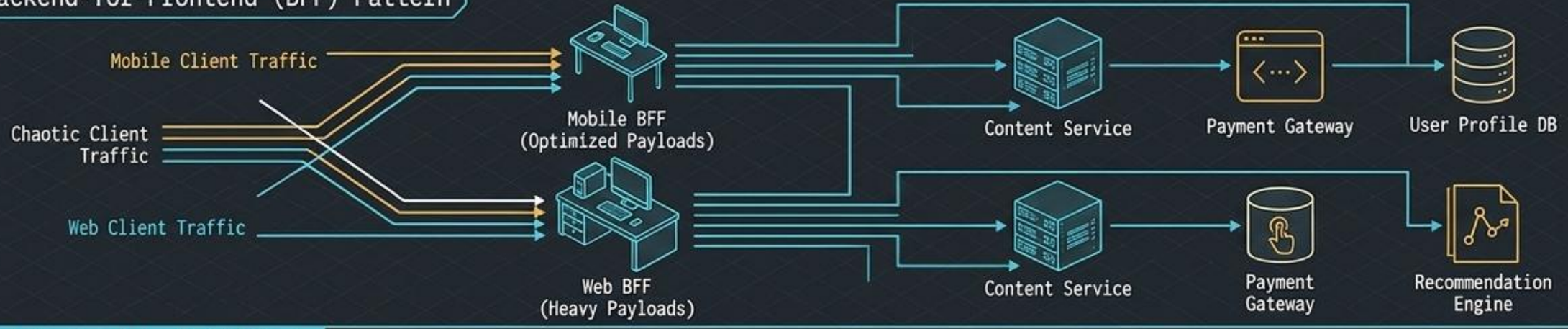
The Traffic Grid: API Gateway & BFF Patterns

Gateways act as intelligent receptionists, funneling chaotic external traffic into structured internal routes.

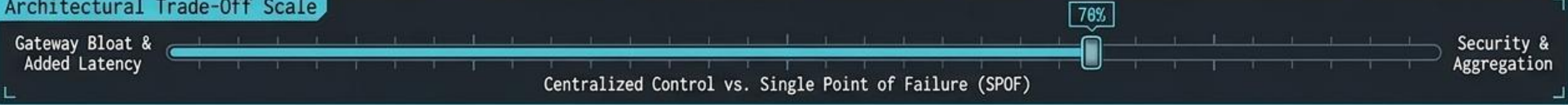
The API Gateway Pattern



Backend-for-Frontend (BFF) Pattern



Architectural Trade-Off Scale



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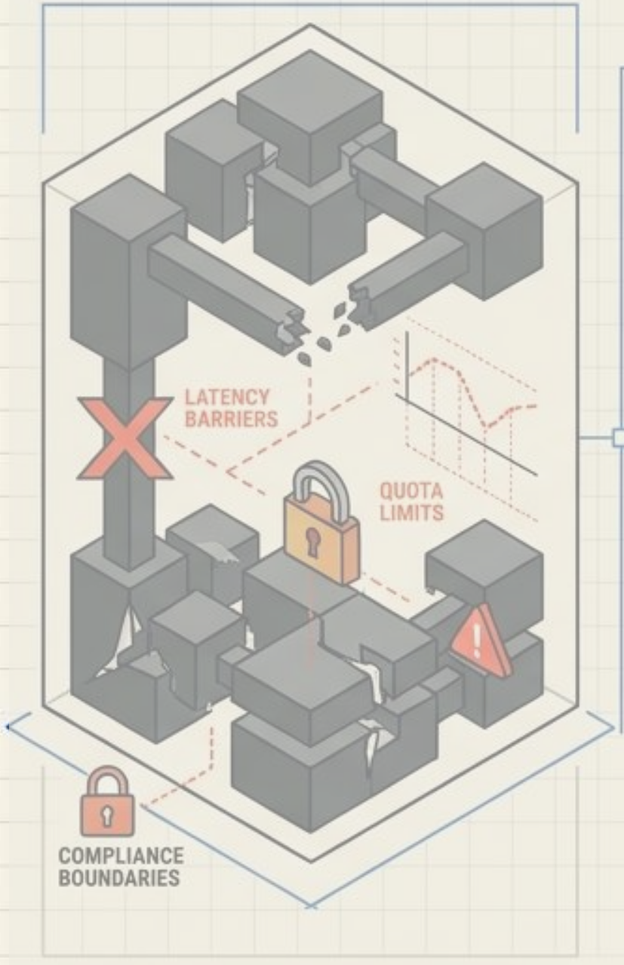

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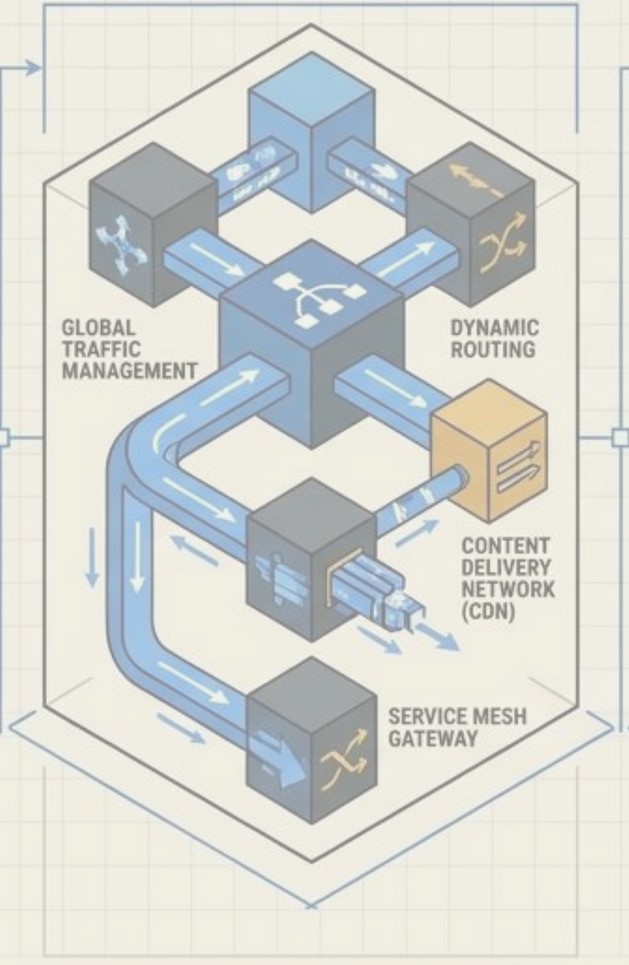
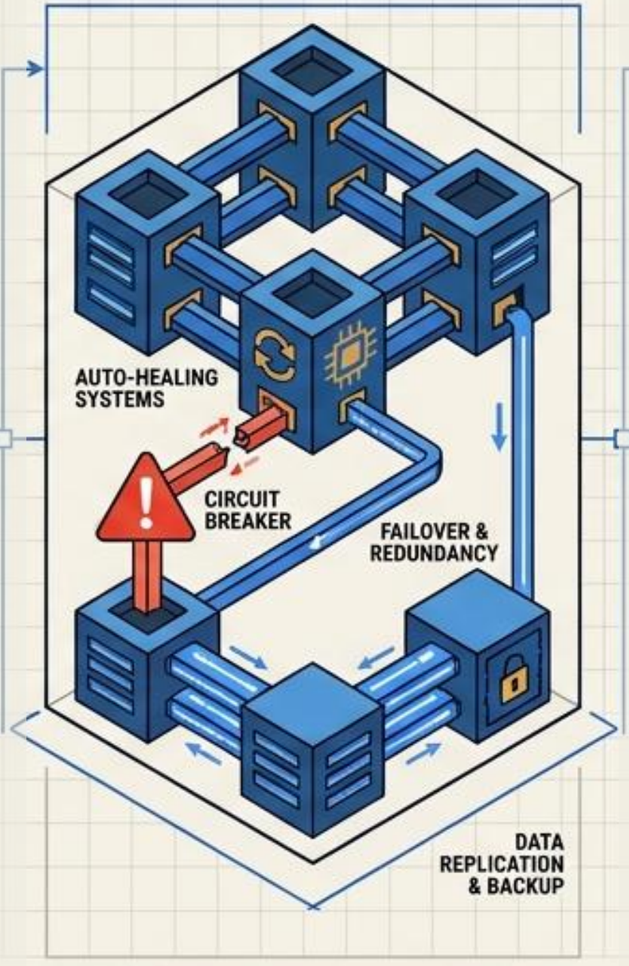
2. Access Routing



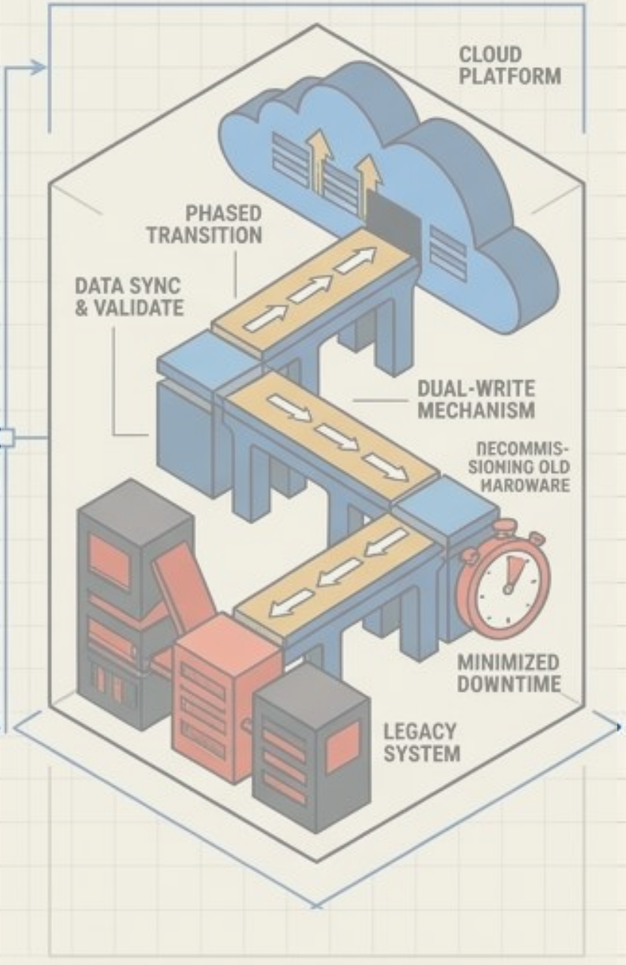
Directing requests efficiently to appropriate resources; optimizes performance and ensures secure entry points.

3. Resiliency Engineering



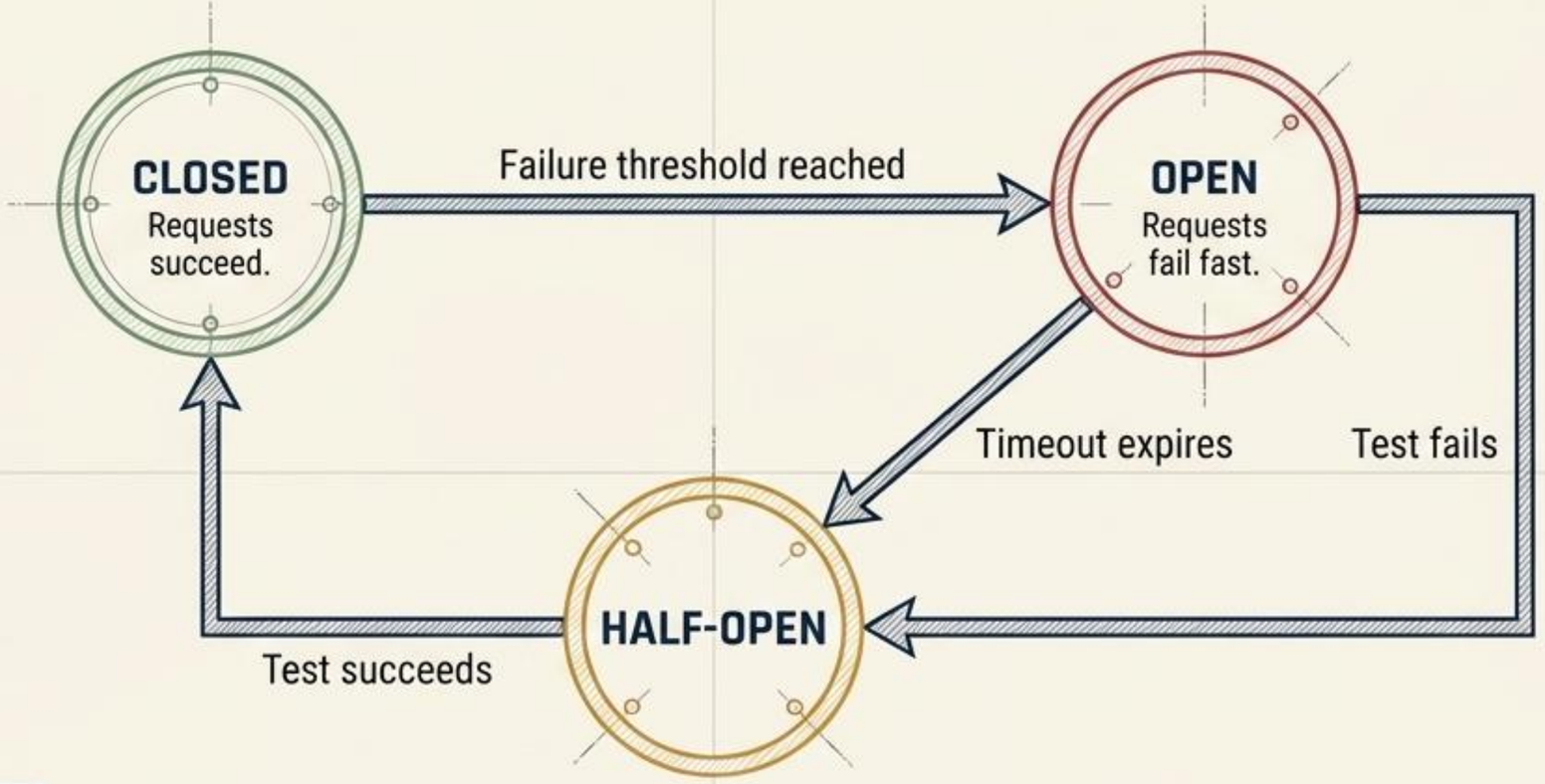
Building fault-tolerant systems that recover from failures; focuses on redundancy and automated recovery strategies.

4. Data & Incremental Migration



Moving data and workloads gradually to the cloud; reduces risk and ensures business continuity during the shift.

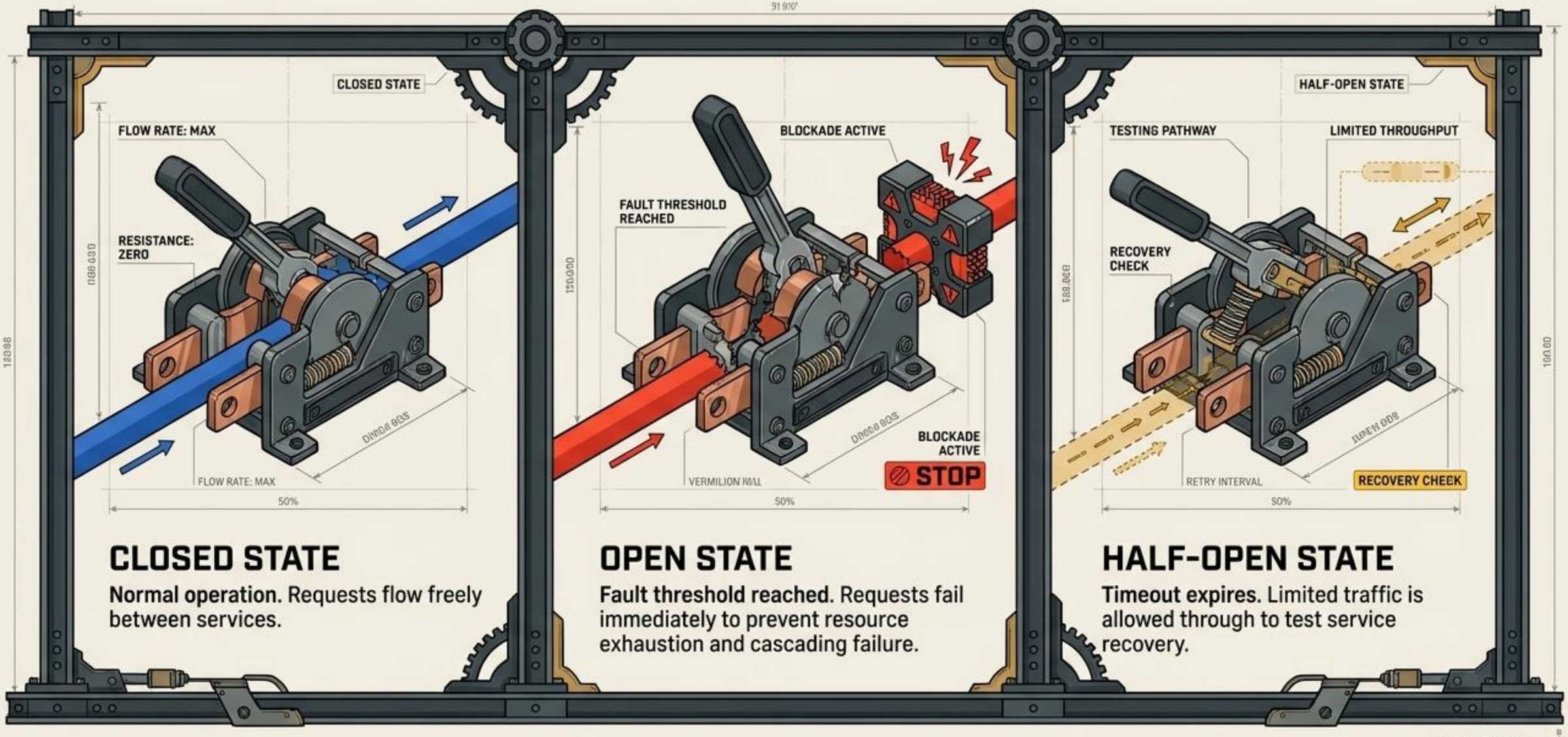
PATTERN 3: CIRCUIT BREAKER (SYSTEM RESILIENCE)



ARCHITECTURAL TRADE-OFFS

	+		−
✓	• Prevents cascading network failures • Gives downstream systems time to recover	−	• Requires complex threshold tuning • UI must be programmed to handle fallback responses

CIRCUIT BREAKERS PREVENT CASCADING SYSTEM FAILURES



CLOSED STATE

Normal operation. Requests flow freely between services.

OPEN STATE

Fault threshold reached. Requests fail immediately to prevent resource exhaustion and cascading failure.

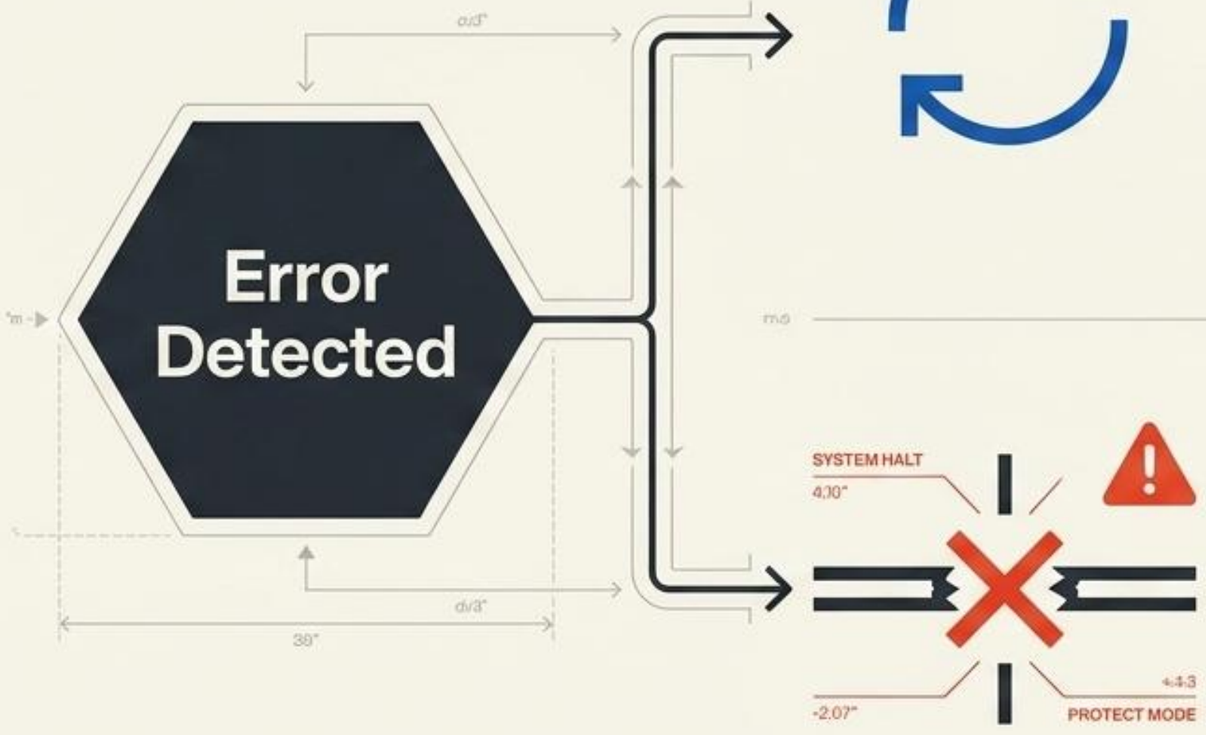
HALF-OPEN STATE

Timeout expires. Limited traffic is allowed through to test service recovery.

ARCHITECTURAL

Match the Resiliency Pattern to the Fault Duration

Circuit Breaker vs. Retry



Retry Pattern

Optimal Use: Transient, brief faults (e.g., a microsecond network blip).

Mechanism: Try again immediately or utilize exponential backoff.

Circuit Breaker Pattern

Optimal Use: Long-lasting faults (e.g., a database crash, severe system overload).

Mechanism: Stop trying. Fail fast to protect surviving infrastructure.

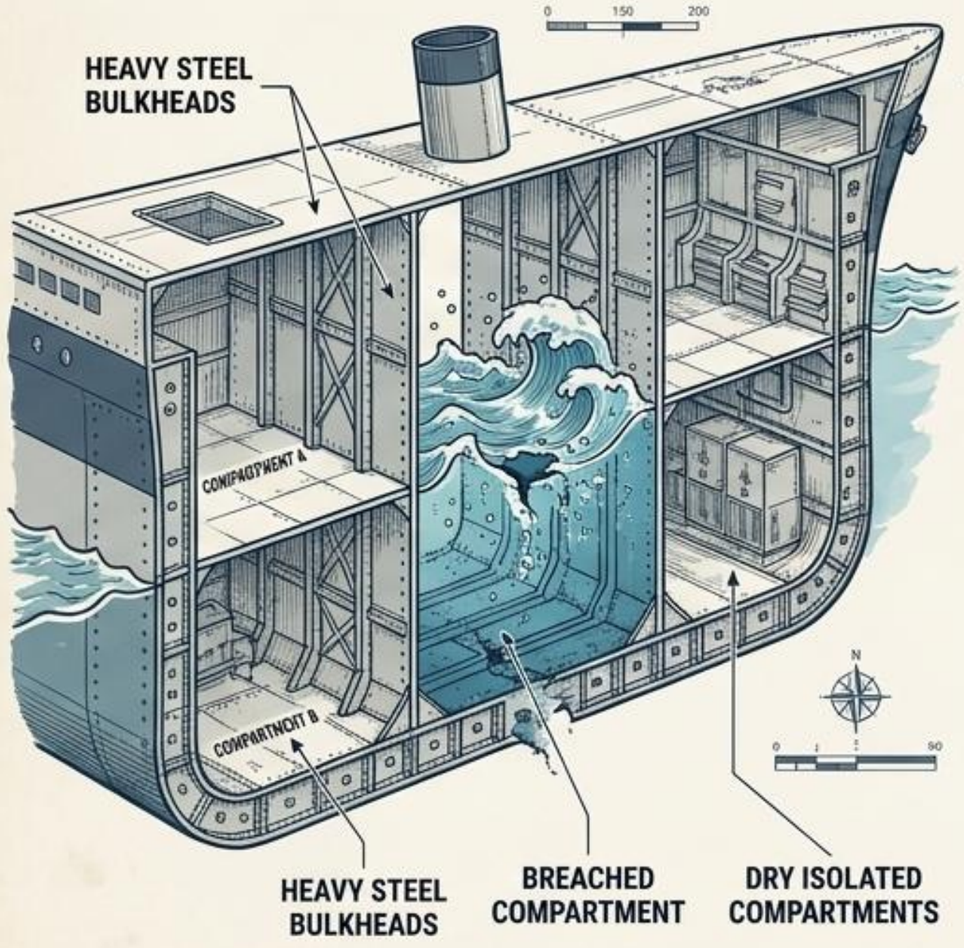
The Resiliency Decision Matrix

Matching the architectural fault to the proper emergency failsafe.

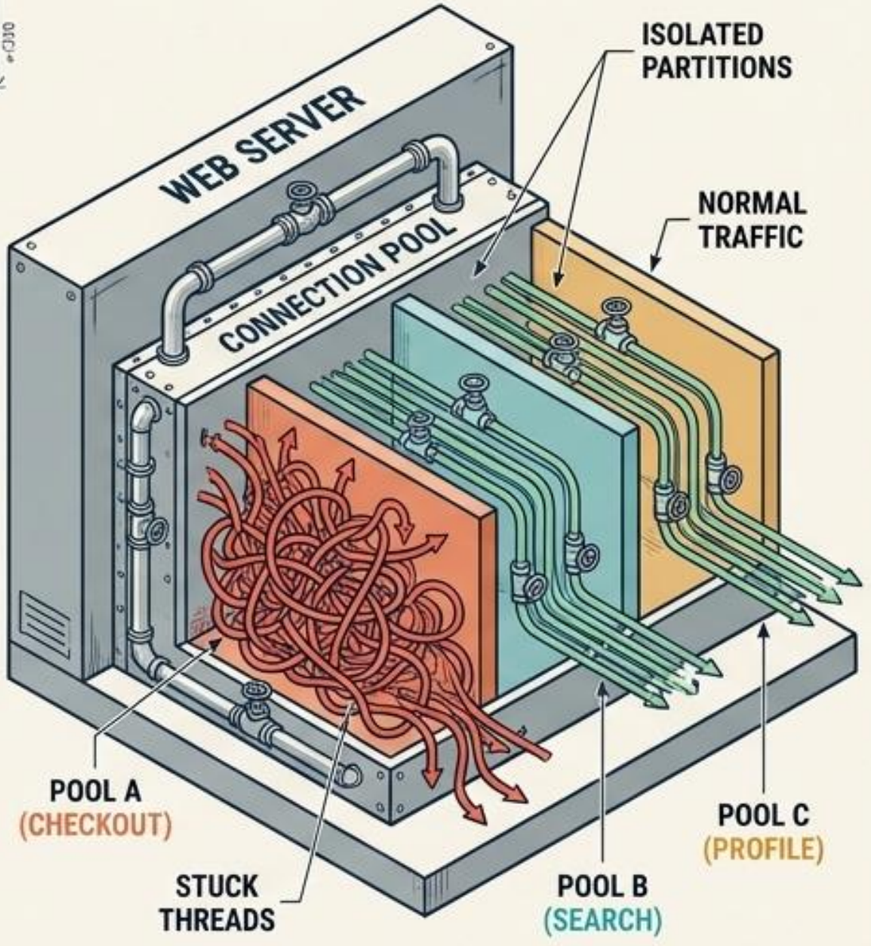
Pattern	Best For (Failure Type)	Response Action	System Impact
Retry Pattern	Transient Network Blips	Exponential Backoff & Re-attempt	Increases downstream load
Circuit Breaker	Systemic DB Outages	Fail Fast / Return Fallback	Relieves downstream load
Bulkhead Pattern	Resource Exhaustion	Isolate & Seal Compartments	Protects core transaction engines
Health Checks	Traffic Routing Control	Remove from Load Balancer	Proactive network stabilization

PATTERN 4: BULKHEAD (RESOURCE ISOLATION)

THE ANALOGY



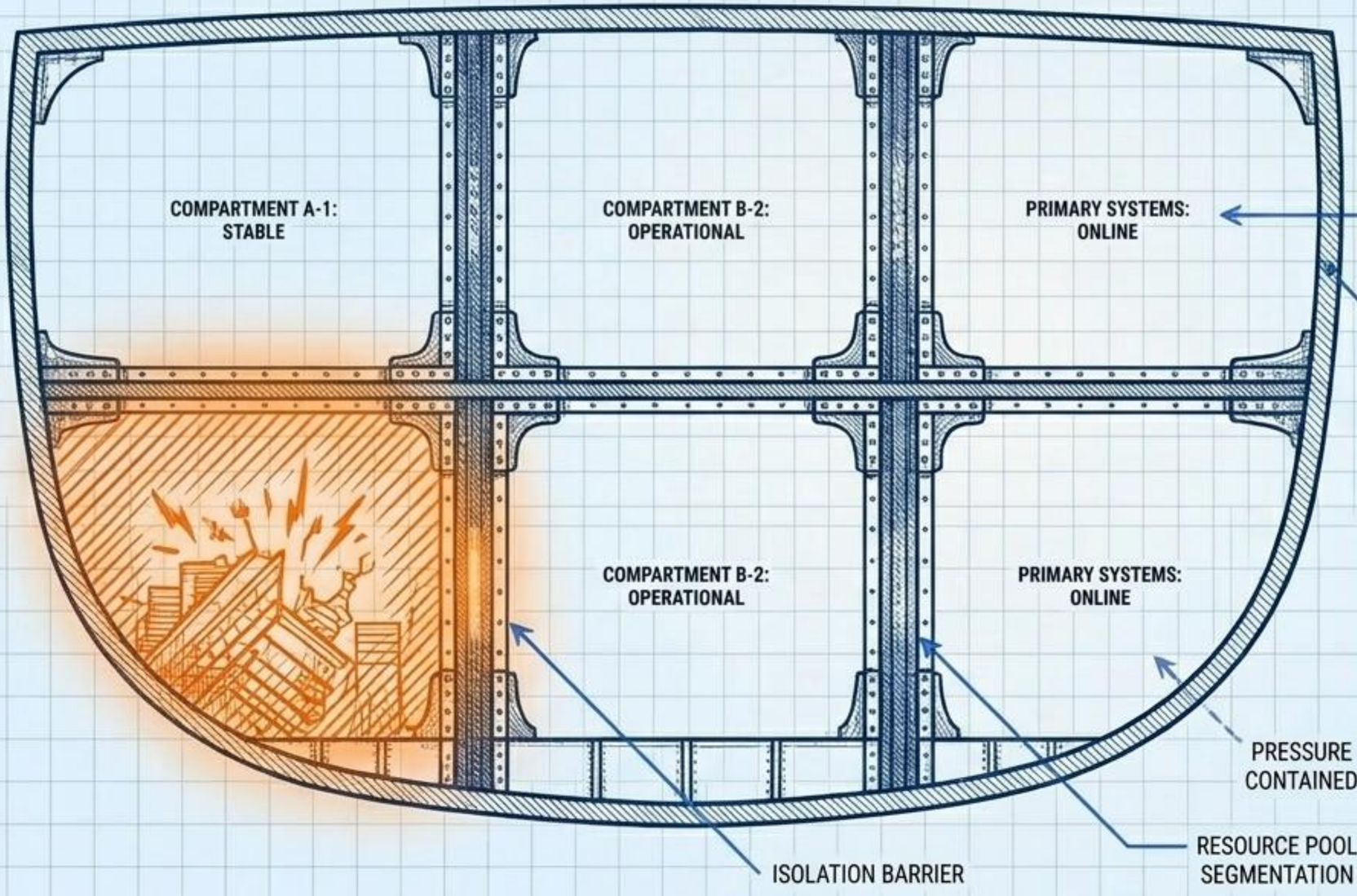
TECHNICAL FLOW



ARCHITECTURAL TRADE-OFFS

+	(+) Exceptional fault isolation
+	(+) Limits blast radius of local failure
+	(+) Predictable performance for critical paths
−	(-) Resource inefficiency (idle thread pools)
−	(-) High configuration complexity

CONTAINING LOCALIZED DAMAGE WITH BULKHEADS



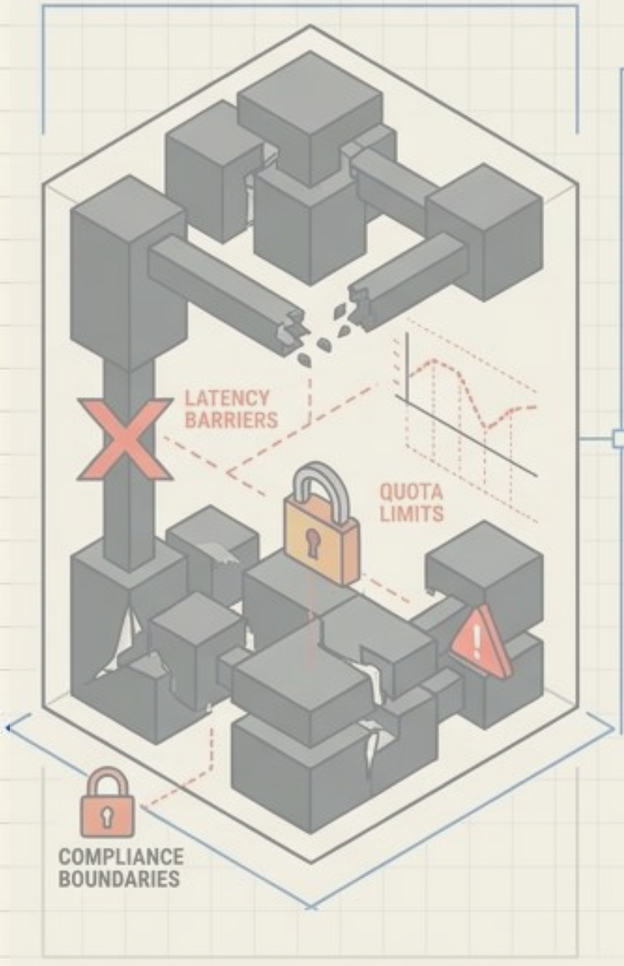
Resource Partitioning Protocol:

- Partitions critical system resources (e.g., threads, memory, connection pools).
- Isolates failures to a single segment or service container.
- Prevents localized spikes from triggering total resource exhaustion.

Cloud Architecture Patterns

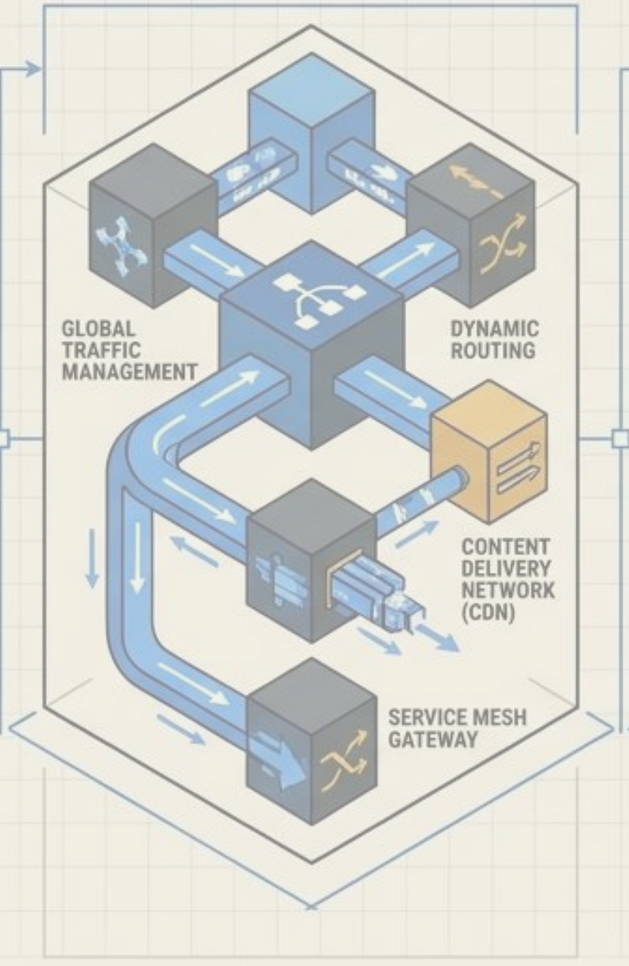
Engineering Resilience and Scale in Distributed Systems

1. Cloud Constraints



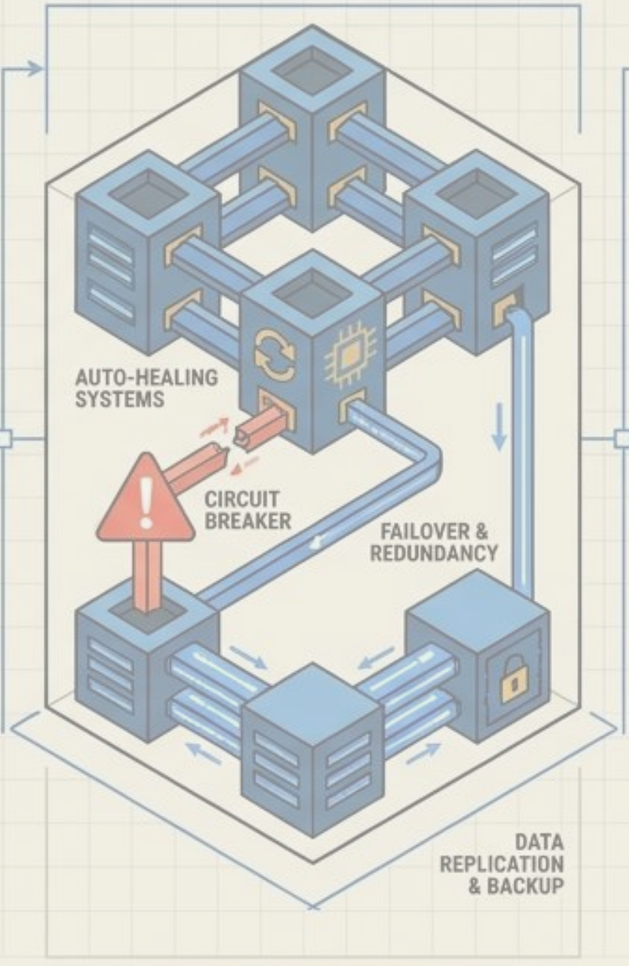
Limitations impacting system design; requires careful consideration of resource caps and geographical restrictions.

2. Access Routing



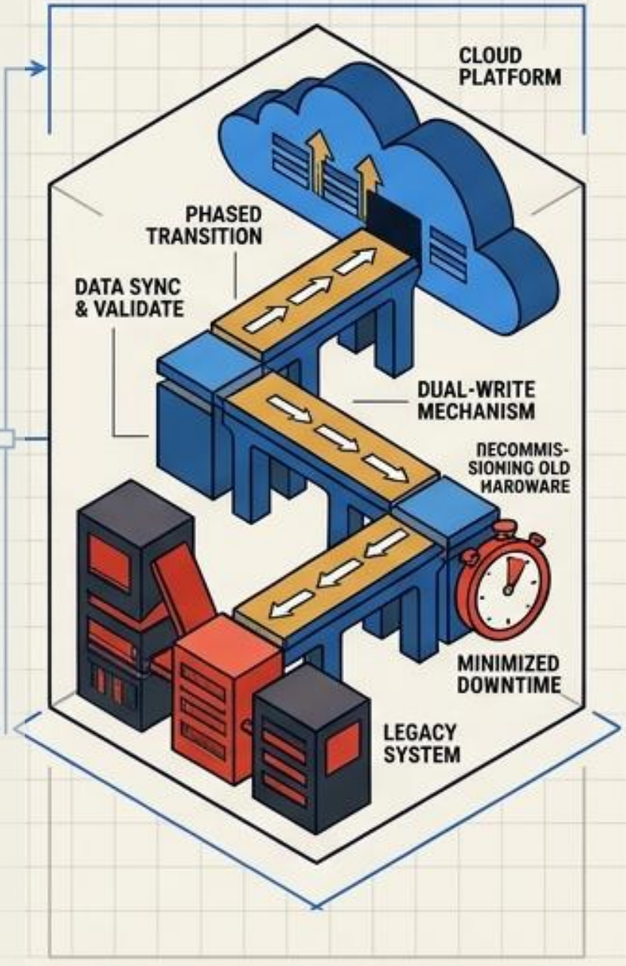
Directing requests efficiently to appropriate resources; optimizes performance and ensures secure entry points.

3. Resiliency Engineering



Building fault-tolerant systems that recover from failures; focuses on redundancy and automated recovery strategies.

4. Data & Incremental Migration

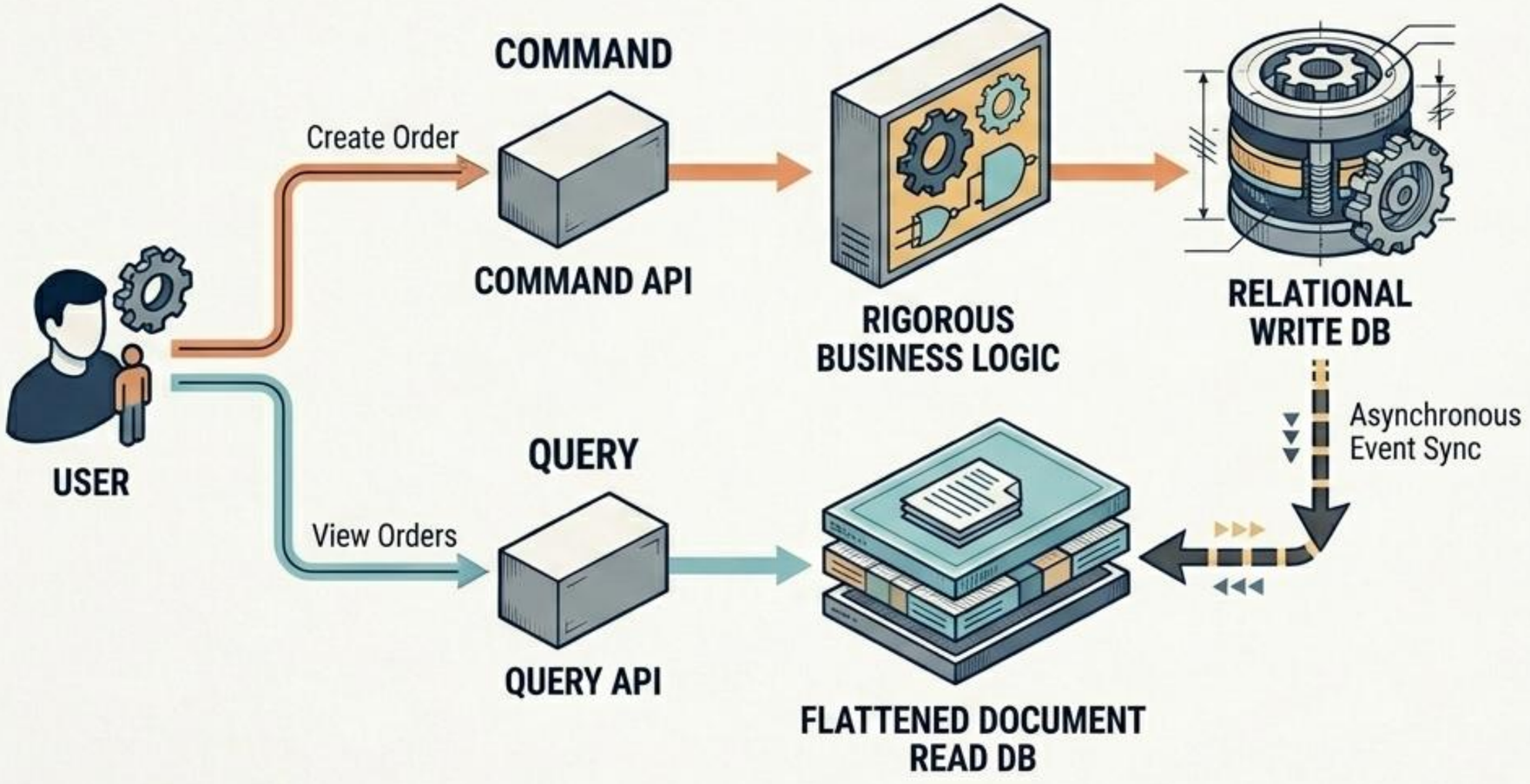


Moving data and workloads gradually to the cloud; reduces risk and ensures business continuity during the shift.

PATTERN 5: CQRS (CLOUD-NATIVE STORAGE)

Command Query Responsibility Segregation

TECHNICAL FLOW



ARCHITECTURAL TRADE-OFFS

⊕	(+) Independent read/write scaling
⊕	(+) Highly optimized database schemas
⊖	(−) Introduces Eventual Consistency
⊖	(−) Massively increases system complexity

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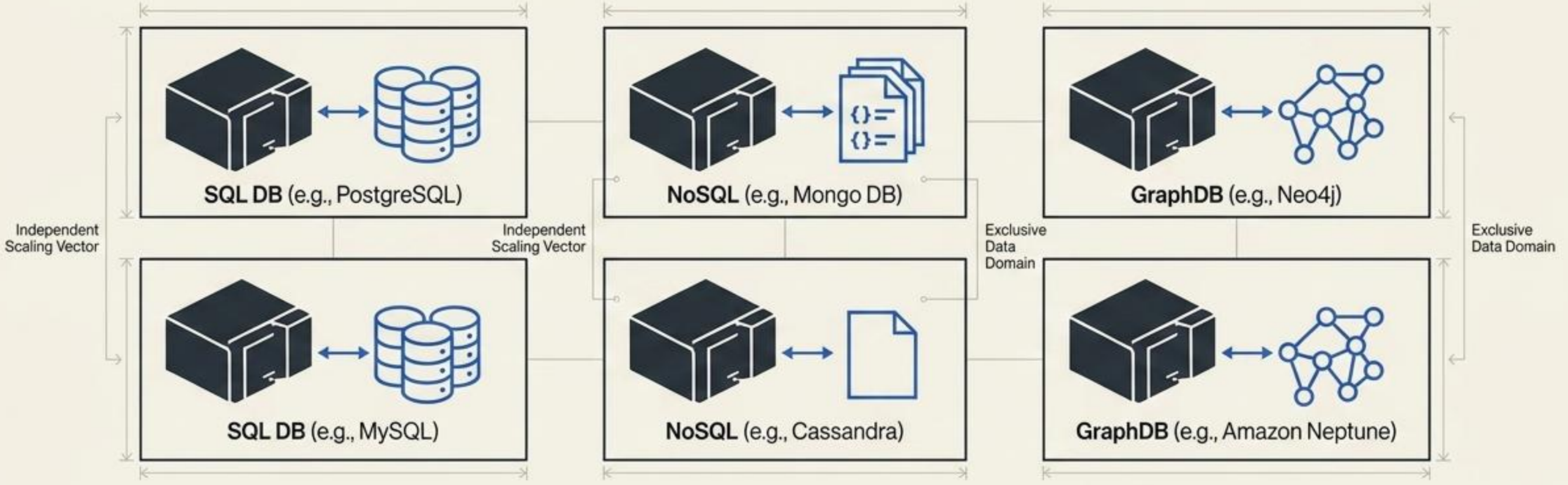
Database-per-Service Enables True Independent Scaling

The Pattern: Each microservice owns its domain data exclusively.



The Benefits

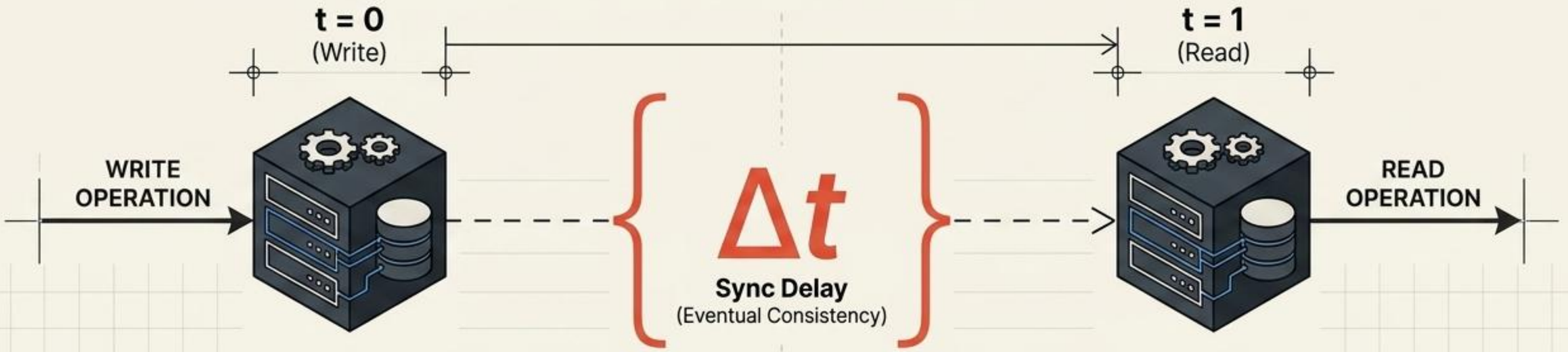
- Uncompromising loose coupling.
- **Polyglot Persistence** (the right database type for the right job).
- Independent horizontal scaling.

The Catch

- **The Catch:** Introduces the massive complexity of Distributed Transactions.

Highly Scalable Reads Require Embracing Eventual Consistency

The architectural trade-off for massive scale read optimization.



The Trade-off

Read models inevitably lag behind write models.

Stale Data

Users might temporarily see outdated information.

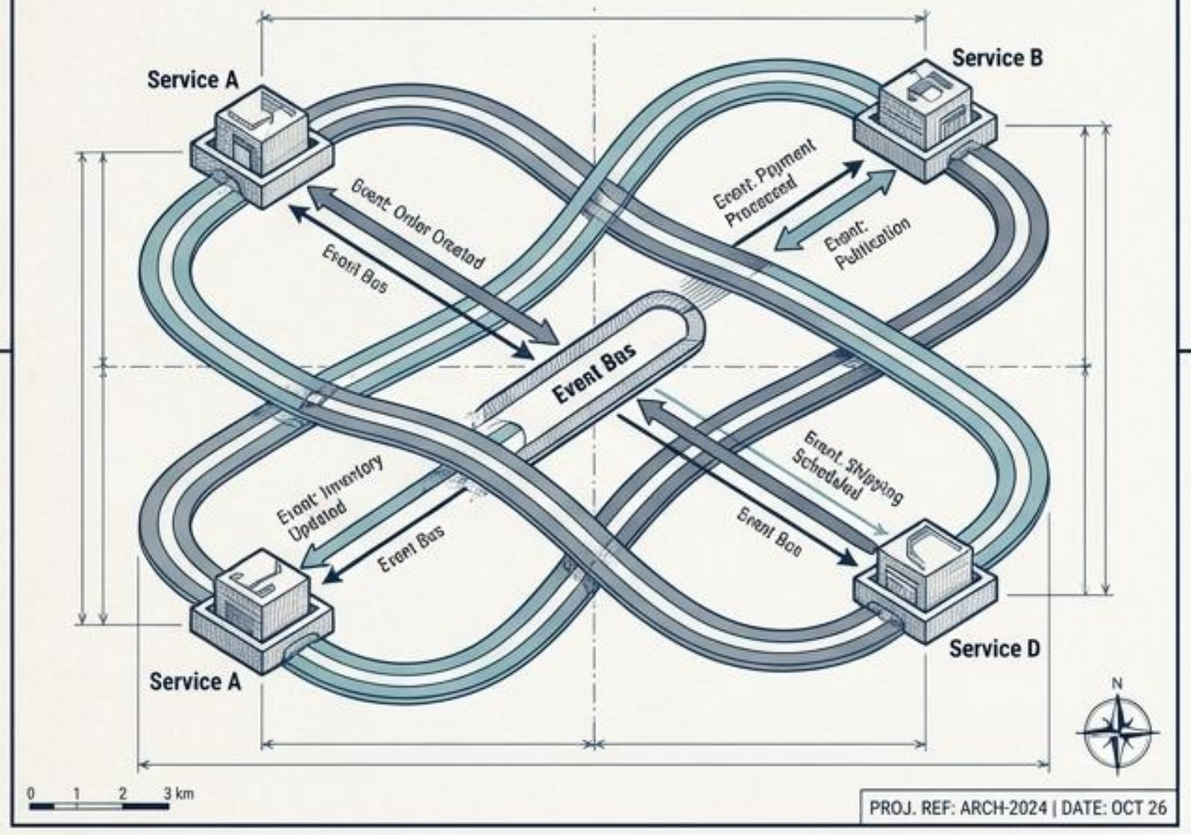
The Solution

Mitigate via UI/UX design (e.g., optimistic frontend updates) rather than relying on slow backend database locking.

PATTERN 6: SAGA (DISTRIBUTED TRANSACTIONS)

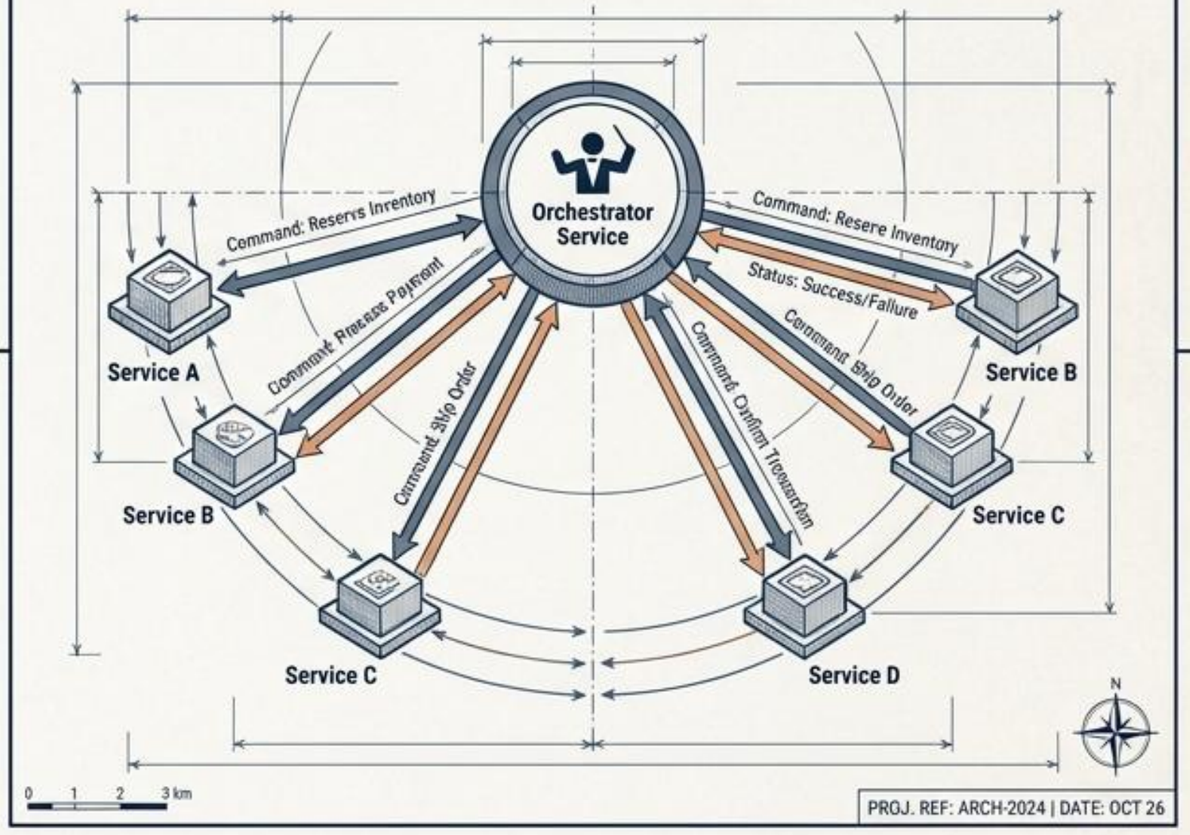
Managing transactions across microservices without locking databases.

CHOREOGRAPHY



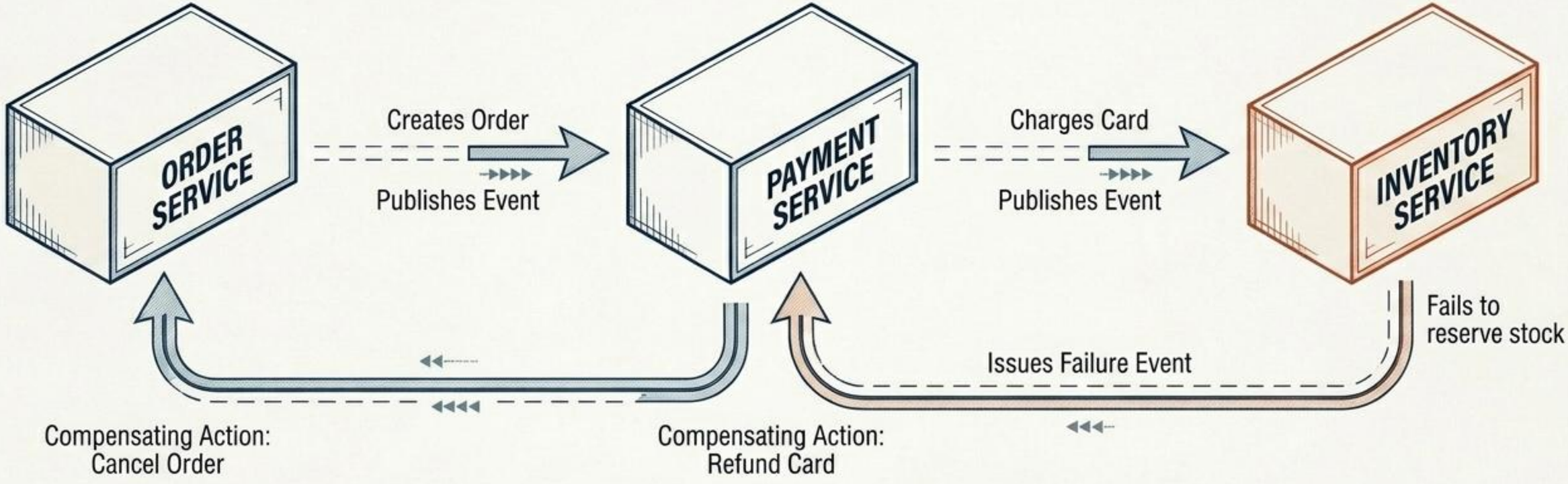
Services act independently, reacting to events published on a shared Event Bus. Like a ballet, there is no central director; each service knows its own routine and reacts to cues.

ORCHESTRATION



A central Orchestrator Service acts as the conductor. It explicitly commands each microservice when to execute its local transaction and tracks the overall state.

THE SAGA FLOW & COMPENSATING TRANSACTIONS



ARCHITECTURAL TRADE-OFFS

+	Maintains data consistency without database locks	−	Extremely complex distributed debugging
+	Highly scalable and fault-tolerant	−	Undo logic must be manually written for every failure state

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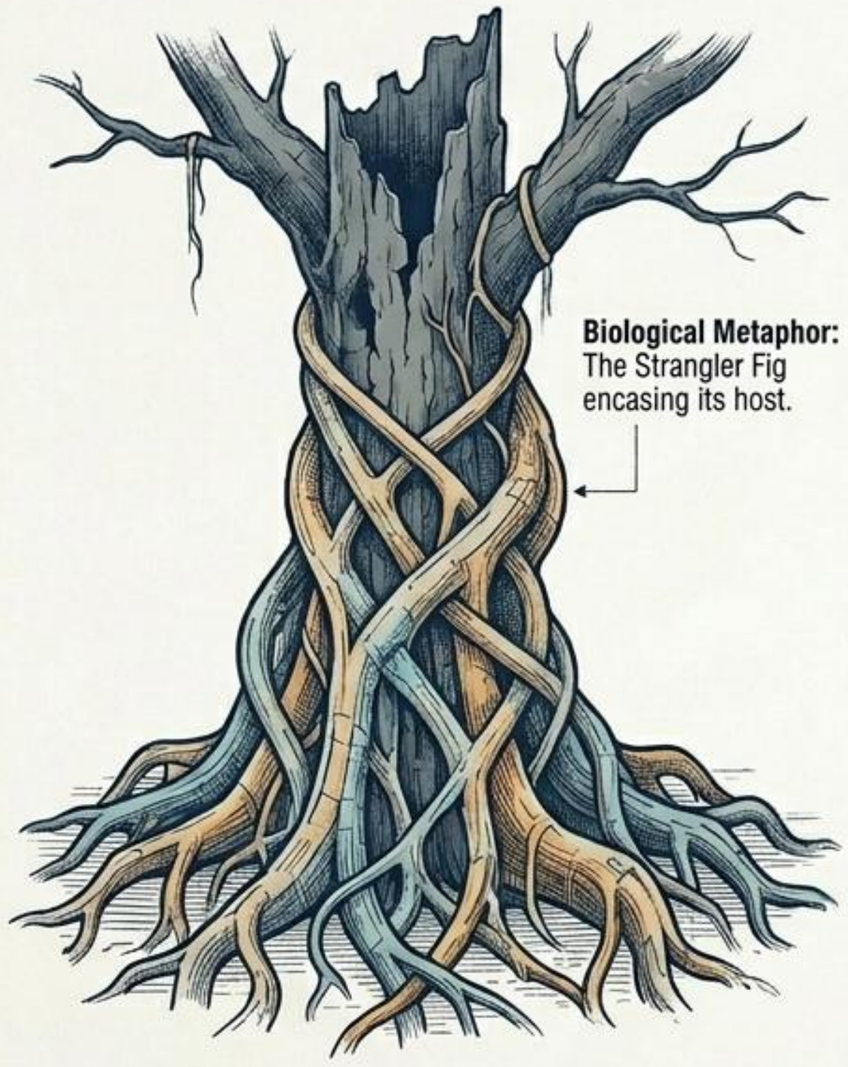
PATTERN 7: STRANGLER FIG (MIGRATION)

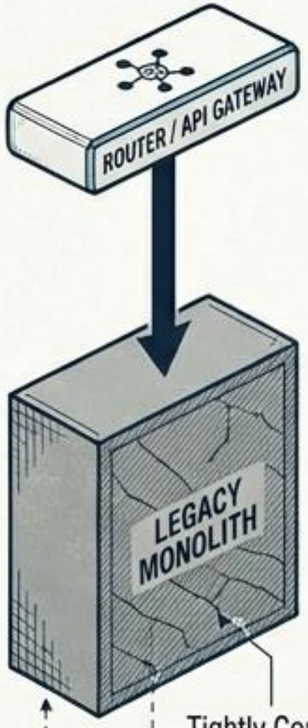
Gradually replacing legacy systems by incrementally migrating functionality to new applications.



Biological Metaphor:
The Strangler Fig encasing its host.

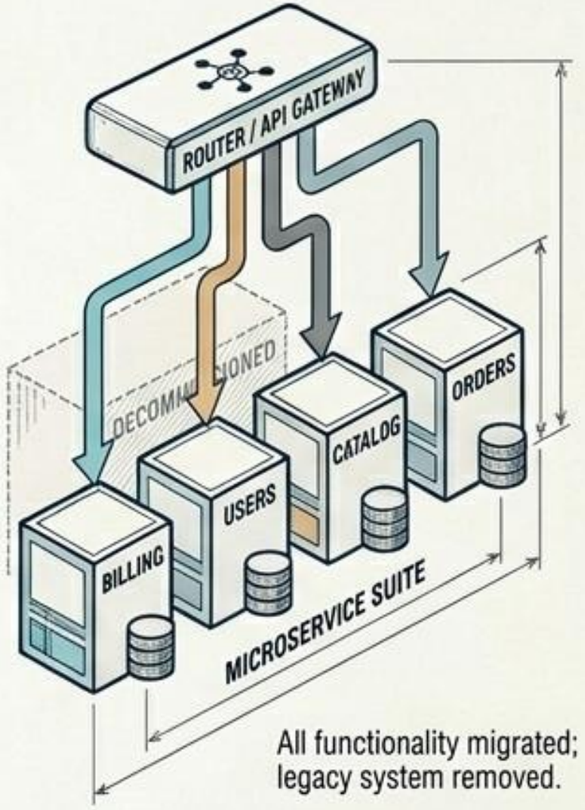
ITERATIVE MODERNIZATION LIFECYCLE

CURRENT STATE



STRANGLING PHASE

REPLACED



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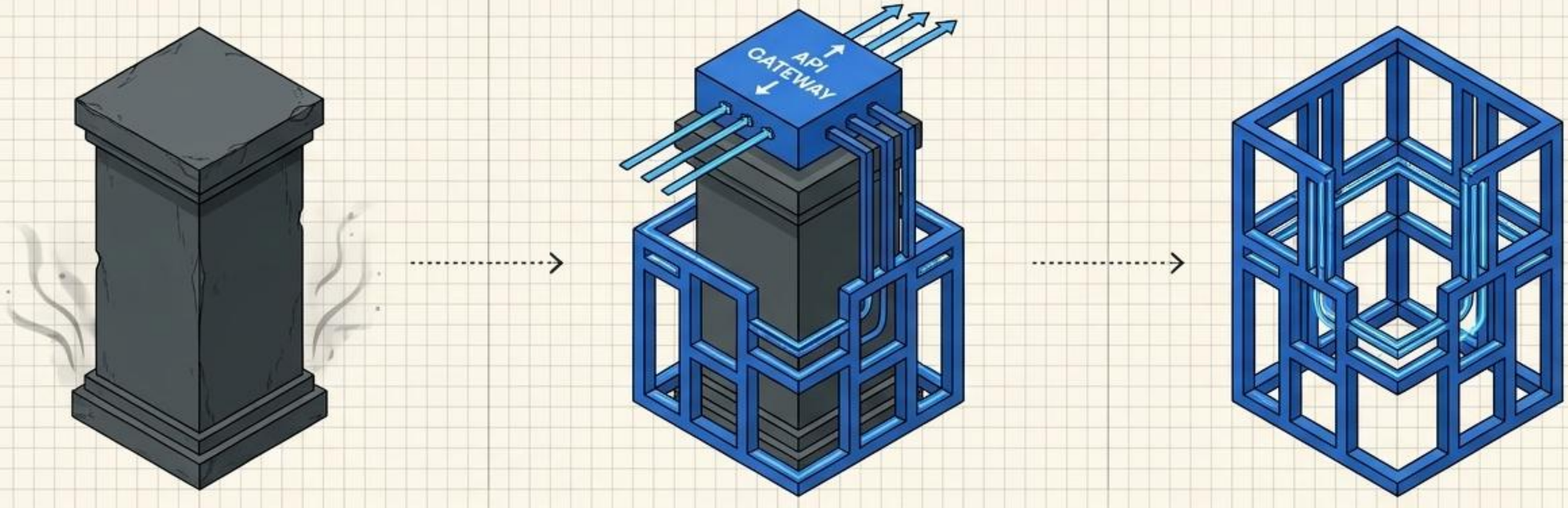
The Strangler Fig Pattern Evolves Systems Incrementally

The Metaphor: A vine that grows around a host tree, eventually replacing it entirely.



Step 1 (Legacy)

Step 2 (Intercept & Route)

Step 3 (Replaced)

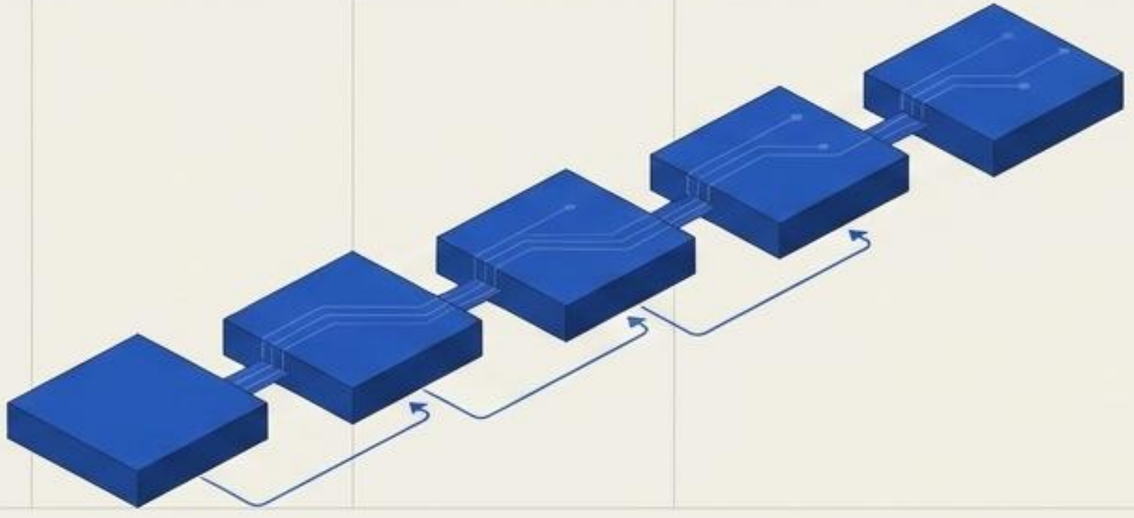
The Strategy:

1. Place an **intercept point** (API Gateway) in front of the legacy monolith.
2. **Gradually route** specific features to newly built microservices.
3. **Safely decommission** the monolith once all traffic is rerouted.

Big Bang Rewrites Introduce Unacceptable Business Risk

The Big Bang Rewrite

Attempting to shut down the old system to launch the completely new system all at once. Extremely high risk of catastrophic failure.

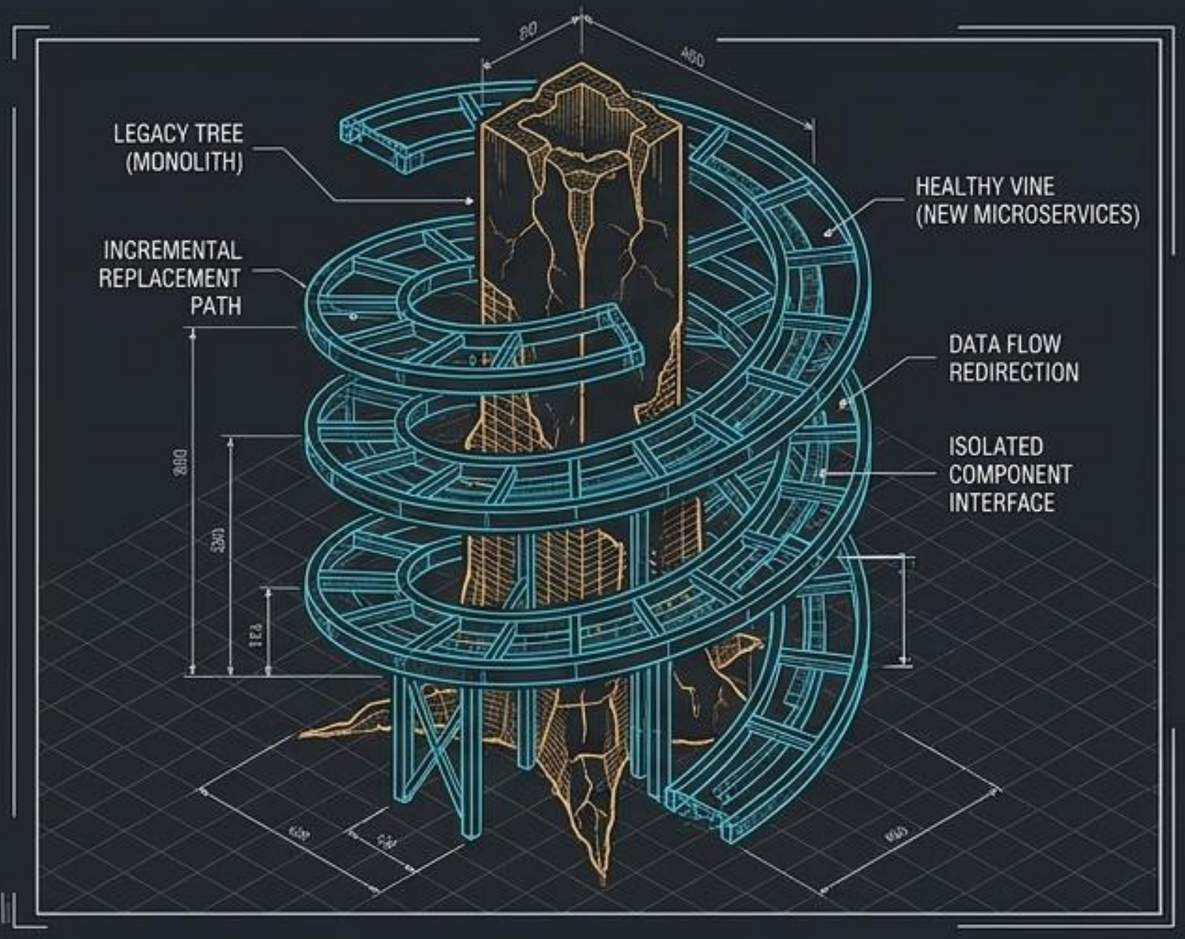


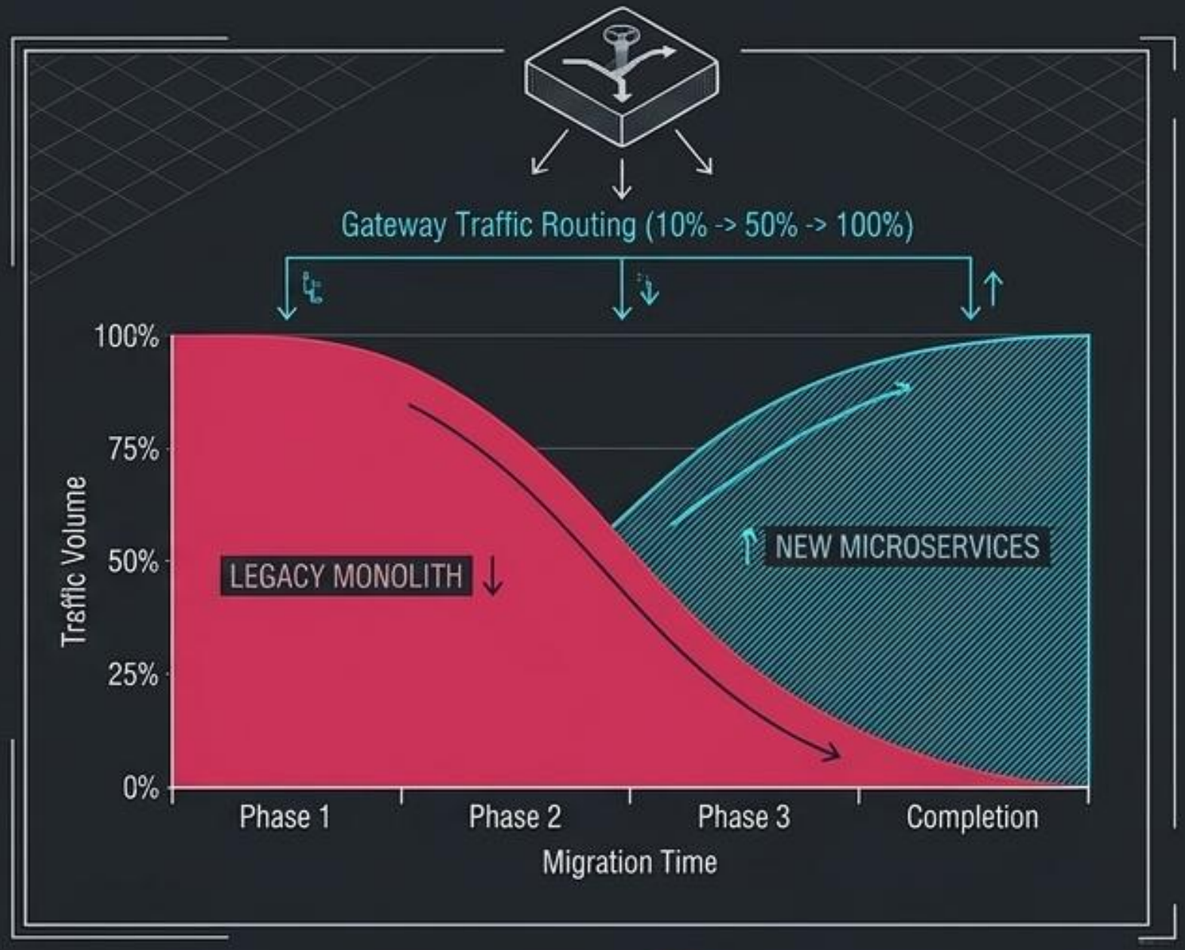
Incremental Migration

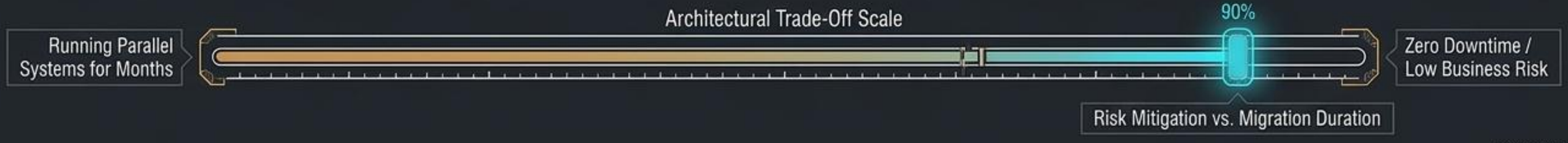
Phased, piece-by-piece transition. Low risk, delivering continuous business value throughout the process.

Urban Renewal: The Strangler Fig Pattern

Never rewrite the monolith; strangle it incrementally behind an API Gateway.



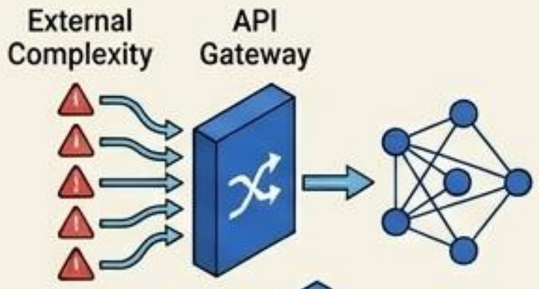


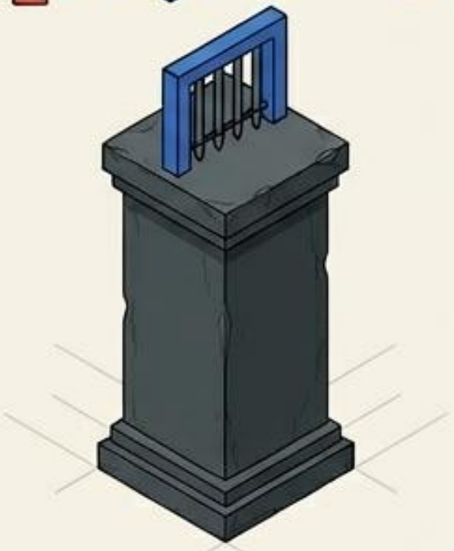


Diagnostic Matrix: Choosing a Migration Path

Strategy	Risk Level	Time to Value	Cloud-Native Benefit
Lift and Shift	Low	Fast	Low (Running legacy code on modern servers)
Big Bang Rewrite	Extreme	Very Slow	High (If launch succeeds)
Strangler Fig	Moderate	Continuous	High (Iterative, safe modernization)

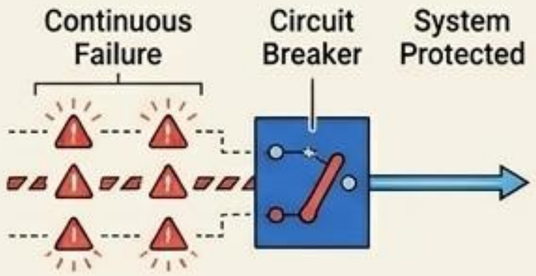
Cloud Architecture Demands Intentional Design for Failure and Scale

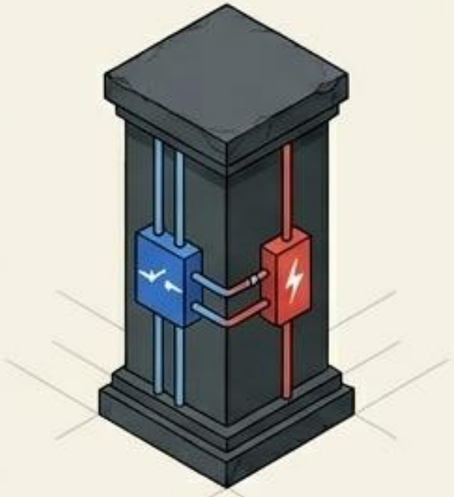




Controlled Access

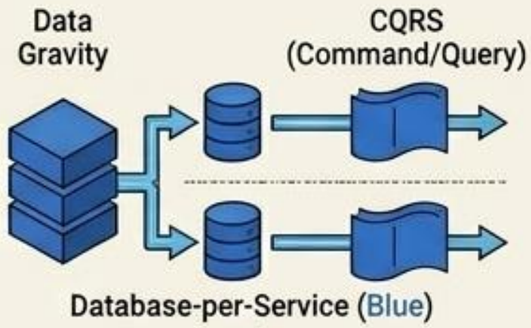
Buffer external complexity and protect internal networks with **API Gateways**.

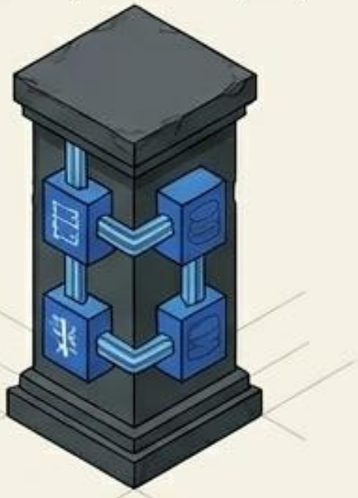




Resilience

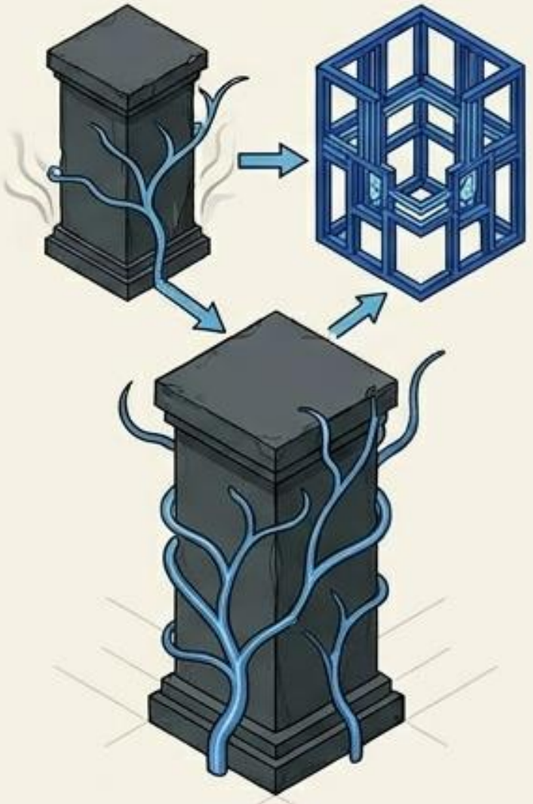
Expect continuous failure; protect the system with **Circuit Breakers**.





Decoupled Data

Break data gravity utilizing **Database-per-Service** and **CQRS**.

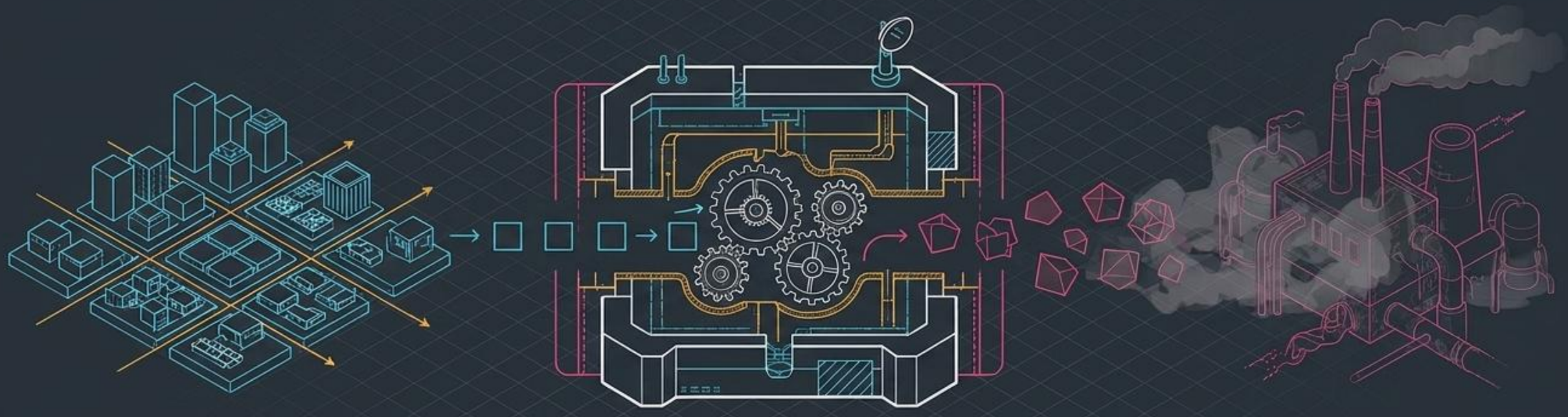


Incremental Evolution

Migrate legacy systems safely via the **Strangler Fig** pattern.

The Anti-Corruption Layer (ACL)

Diplomatic translators protecting cloud-native domains from legacy technical debt.



Modern Microservices
(Clean Domain Driven Design)

Anti-Corruption
Translation Layer

Legacy Mainframe
(Archaic Data Structures)

Architectural Trade-Off Scale' UI component

75%

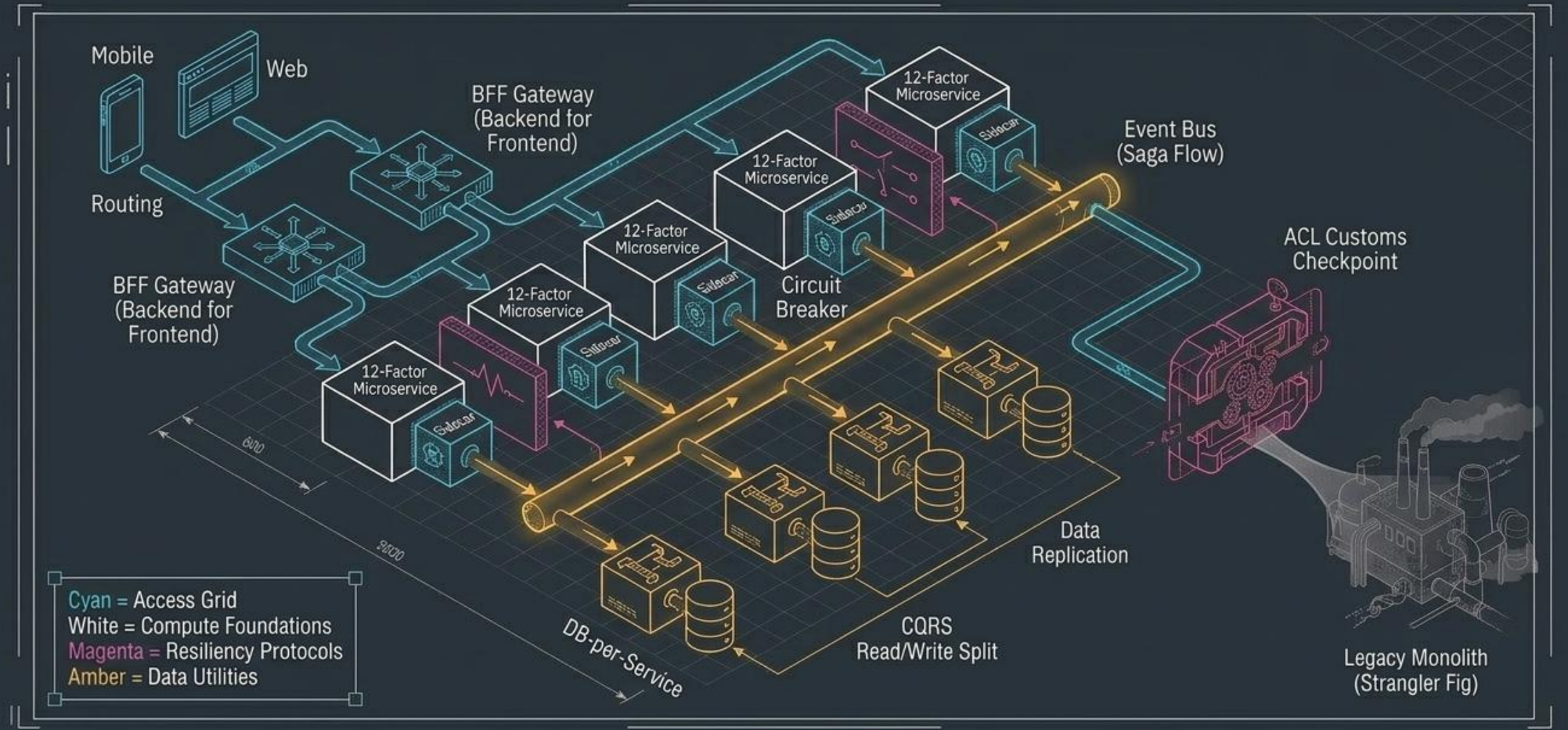
Added Latency &
Development Overhead

Zero Technical
Debt Leakage

Domain Purity vs. Translation Overhead

The Master Blueprint: The Complete Metropolis

A resilient, decoupled, and evolving ecosystem that survives scale and outlives its legacy origins.



There are no perfect architectures, only optimized trade-offs.

Design for inevitable failure.

Decouple to scale independently.

Migrate iteratively, never all at once.

Based on *Cloud Application Architecture Patterns* by Brown, Woolf, and Yoder